Matrix Decomposition Theorem

Matrix **decomposition** is a factorization of a matrix into some canonical form (Standard form/Normal form). There are many different matrix decompositions; each finds use among a particular class of problems.

Decompositions related to solving systems of linear equations

- LU decomposition
- Cholesky decomposition
- QR decomposition

Decompositions based on eigenvalues and related concepts

- **Eigen decomposition (also called spectral/diagonal decomposition)**
- Singular value decomposition

A. Eigen/diagonal Decomposition or spectral decomposition Theorem:

In linear algebra, eigendecomposition or sometimes spectral decomposition is the factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors. Only diagonalizable matrices can be factorized in this way.

Let A be the given Matrix, D is a diagonal matrix with the corresponding eigen values of Matrix A and **P be a matrix of eigenvectors** of a given Matrix A, Then A can be written as an eigen decomposition $\mathbf{A = PDP^{-1}}$

Example 67: Examine the eigen decomposition for the matrix $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

Let $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$

$$\therefore \mathbf{A} - \lambda \mathbf{I} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{pmatrix}$$

$$|\mathbf{A} - \lambda \mathbf{I}| = \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1$$

$$\Rightarrow \quad |\mathbf{A} - \lambda \mathbf{I}| = 0$$

$$\Rightarrow \quad \begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} = (4-\lambda)(4-\lambda) - 1 = 0$$

$$\Rightarrow (16 - 4\lambda - 4\lambda + \lambda^2) - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 16 - 1 = 0$$

$$\Rightarrow \lambda^2 - 8\lambda + 15 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda - 3\lambda + 15 = 0$$

$$\Rightarrow \lambda(\lambda - 5) - 3(\lambda - 5) = 0$$

$$\Rightarrow (\lambda-5)(\lambda-3)=0$$
$$\Rightarrow \lambda=5, \lambda=3$$

$$\therefore D=\begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$$

Now to find out **the eigenvector** of the matrix A,

Let a non-zero vector v = $\begin{bmatrix} x \\ y \end{bmatrix}$

We have, $\mathbf{Av=\lambda v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=\lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix}=3\begin{pmatrix} x \\ y \end{pmatrix} \quad \text{[For } \lambda=3\text{]}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix}=\begin{pmatrix} 3x \\ 3y \end{pmatrix}$$

$$\Rightarrow 4x+y=3x$$
$$x+4y=3y$$
$$\Rightarrow x+y=0$$
$$x+y=0$$

$$\Rightarrow x+y=0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 then x = -1

$\therefore$ Eigenvector $\mathbf{v}=\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for $\lambda=3$

Again for $\lambda=5$

Let a non-zero vector v = $\begin{bmatrix} x \\ y \end{bmatrix}$

We have, $\mathbf{Av=\lambda v}$

$$\Rightarrow \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}=5\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix}=5\begin{pmatrix} x \\ y \end{pmatrix} \quad \text{[For } \lambda=5\text{]}$$

$$\Rightarrow \begin{pmatrix} 4x+y \\ x+4y \end{pmatrix}=\begin{pmatrix} 5x \\ 5y \end{pmatrix}$$

$$\Rightarrow 4x + y = 5x$$
$$x + 4y = 5y$$

$$\Rightarrow -x + y = 0$$
$$x - y = 0$$

$$\Rightarrow x - y = 0$$
$$x - y = 0$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in **particular (2-1) =1 free variable**, which is y.
Let y =1 , then x =1

$\therefore$ Eigenvector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $\lambda = 5$

P be a matrix of eigenvectors of a given Matrix A.

$$\therefore \mathbf{P} = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$$

Now to find out $\mathbf{p}^{-1}$

$$c_{11} = 1,\ c_{12} = -1,\ c_{21} = -1,\ c_{22} = -1$$

$$p^c = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$\text{Adj}\,P = (P^C)^T = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

$$|P| = \begin{vmatrix} -1 & 1 \\ 1 & 1 \end{vmatrix} = -1 - 1 = -2$$

$$\therefore P^{-1} = \frac{\text{Adj}P}{|P|} = -\frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

Then A can be written as an eigen decomposition $\mathbf{A = PDP^{-1}}$

Justification:

$$\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$
$$= -\frac{1}{2}\begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 3 & -3 \\ -5 & -5 \end{pmatrix}$$
$$= -\frac{1}{2}\begin{pmatrix} -3-5 & 3-5 \\ 3-5 & -3-5 \end{pmatrix}$$

$$= -\frac{1}{2}\begin{pmatrix} -8 & -2 \\ -2 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix} = \mathbf{A} \quad \text{(Proved)}$$

$\mathbf{A = PDP^{-1}}$ (Proved)

Home task: Find eigen decomposition of the matrix $= \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$ and $= \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

Q-01: What is the difference between Eigen decomposition and SVD?

The SVD always exists for any sort of rectangular or square matrix, whereas the eigen decomposition can only exists for square matrices, and even among square matrices sometimes it doesn't exist.

What is singular value of a Matrix?

The singular values of a $\mathbf{m \times n}$ matrix A are the square roots of the eigenvalues of the symmetric $\mathbf{n \times n}$ matrix $\mathbf{A^T A}$. The Eigen values of $\mathbf{A^T A}$ are

$\boldsymbol{\lambda_1 \geq \lambda_2 \ldots\ldots\ldots\ldots \geq \lambda_r > \lambda_{r+1} \ldots\ldots\ldots\ldots \geq \lambda_n}$.

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

$\boldsymbol{\sigma_i = \sqrt{\lambda_i}}$ are called the singular value of A where i = 1, 2,……………, n.

It is customary to denote the singular values by $\boldsymbol{\sigma_1, \sigma_2, \sigma_3, \ldots\ldots\ldots\sigma_n}$ and to list them in decreasing order:

$\boldsymbol{\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \ldots\ldots\ldots \geq \sigma_n}$

Singular Value Decomposition:

In linear algebra, the **singular value decomposition (SVD)** is an important factorization of a rectangular real or complex matrix, with several applications **in signal processing , image processing and compression, gene expression analysis and statistics**. Applications which employ the SVD include computing the pseudo-inverse, matrix approximation, and determining the rank, range and null space of a matrix.

The spectral theorem says that normal matrices, which by definition must be square, can be unitarily diagonalized using a basis of eigenvectors. The SVD can be seen as a generalization of the spectral theorem to arbitrary, not necessarily square, matrices.

The singular value decomposition (SVD) is a generalization of the eigen-decomposition, which can be used to analyze rectangular matrices (the eigen- decomposition is defined only for squared matrices). By analogy with the eigen-decomposition, which decomposes a matrix into two simple matrices, the main idea of the SVD is to decompose a rectangular matrix into three simple matrices: Two orthogonal matrices and one diagonal matrix.

Matrix A is decomposed using SVD to acquire an orthogonal base in the multidimensional feature space:

$$\underbrace{\begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} * & \\ & * \\ & \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} * & * \\ * & * \end{bmatrix}}_{V^T}$$

$\mathbf{A = USV^T}$

Here,

$\mathbf{A \rightarrow}$ **Given Matrix**

$\mathbf{U \rightarrow}$ **Matrix of Eigen Vectors of the Matrix** $\mathbf{AA^T}$

$\mathbf{V \rightarrow}$ **Matrix of Eigen Vectors of the Matrix** $\mathbf{A^TA}$

$\mathbf{S \rightarrow}$ **Roots of Eigen Values of the Matrix** $\mathbf{AA^T}$ **or** $\mathbf{A^TA}$

$\mathbf{V^T \rightarrow}$ **Transpose of the Matrix** $\mathbf{V}$

Example 68: Compute Singular Value Decomposition (SVD) for the given matrix $\begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

Answer: Let, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

In order to find U, Now we need to find the eigenvectors for AA^T

Given, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\therefore \mathbf{A^T} = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

So, $\mathbf{AA^T} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 4+4 & 2-2 \\ 2-2 & 1+1 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \qquad \text{--------------------------------(i)}$$

Now we need to find the eigenvalues of $\mathbf{AA^T}$

$$AA^T - \lambda I = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$
$$= \begin{pmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{pmatrix}$$

$$\left|AA^T - \lambda I\right| = 0$$
$$\Rightarrow \begin{vmatrix} 8-\lambda & 0 \\ 0 & 2-\lambda \end{vmatrix} = 0$$
$$\Rightarrow (8-\lambda)(2-\lambda) = 0$$
$$\Rightarrow (\lambda-8)(\lambda-2) = 0$$
$$\Rightarrow \lambda = 8,2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $S = \sqrt{D} = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$

$$S = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{--------------------------(ii)}$$

Now we need to find the eigenvector of AA^T

We have, $Au_1 = \lambda u_1$ $[AV = \lambda V]$

Let $u_1 = \begin{pmatrix} x \\ y \end{pmatrix}$

$\therefore AA^T u_1 = \lambda u_1$

$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 8\begin{pmatrix} x \\ y \end{pmatrix}$ [For $\lambda = 8$ & from (i) putting the value of AA^T]

$$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$
$$\Rightarrow 8x = 8x$$
$$2y = 8y$$

$\Rightarrow 8x - 8x = 0$

$2y - 8y = 0$

$\Rightarrow 8x = 8x$

$-6y = 0$

$\Rightarrow 8x = 8x$

$y = 0$

$\therefore y = 0$

Let $x = -1$

$\therefore u_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$ It is a unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_1}| = \sqrt{(-1)^2 + (0)^2} = \sqrt{1} = 1$

Again, Let $\mathbf{u_2} = \begin{pmatrix} x \\ y \end{pmatrix}$

Here,

$\therefore AA^T u_2 = \lambda u_2$

$\begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix}$ [For $\lambda = 2$ & from (i) putting the value of AA^T]

$\Rightarrow \begin{pmatrix} 8x \\ 2y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$

$\Rightarrow 8x = 2x$

$2y = 2y$

$\Rightarrow 8x - 2x = 0$

$2y = 2y$

$\Rightarrow 6x = 0$

$2y = 2y$

$\therefore x = 0$

$2y = 2y$

Let $y = 1$

$\therefore u_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ It is a unit vector or normal vector. Since the length of $\mathbf{u_1}$ is $|\mathbf{u_2}| = \sqrt{(0)^2 + (1)^2} = \sqrt{1} = 1$

$$\therefore U = \begin{pmatrix} u_1 & u_2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{----------------------------------(iii)}$$

Now we need to find the eigenvectors for $A^T A$

Given, $\mathbf{A} = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$

$$\therefore \mathbf{A}^{\mathbf{T}} = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}$$

$$\therefore \mathbf{A}^{\mathbf{T}}\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} \quad \text{------------------------------------(iv)}$$

Now we need to find the eigenvalues of $\mathrm{A^T A}$

$$\therefore \mathbf{A}^{\mathbf{T}}\mathbf{A} - \lambda\mathbf{I} = \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{pmatrix}$$

$$\therefore \left|\mathbf{A}^{\mathbf{T}}\mathbf{A} - \lambda\mathbf{I}\right| = 0$$

$$\Rightarrow \begin{vmatrix} 5-\lambda & -3 \\ -3 & 5-\lambda \end{vmatrix} = 0$$

$$(5-\lambda)(5-\lambda) - 9 = 0$$

$$\Rightarrow 25 - 10\lambda + \lambda^2 - 9 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda + 16 = 0$$

$$\Rightarrow (\lambda - 8)(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 8,2$$

Therefore eigenvalues are 8 and 2. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

The matrix of singular values $S = \begin{pmatrix} \sqrt{8} & 0 \\ 0 & \sqrt{2} \end{pmatrix}$

$$\mathbf{S} = \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \text{---------------------------(v)}$$

Now we need find the eigenvectors of $A^T A$ which are the columns of V.

We have, $\mathbf{Av_1} = \lambda \mathbf{v_1}$ $\qquad\qquad\qquad\qquad$ $[AV = \lambda V]$

Let $\mathbf{v_1} = \begin{pmatrix} x \\ y \end{pmatrix}$

$\therefore \mathbf{A^T A v_1} = \lambda \mathbf{v_1}$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 8\begin{pmatrix} x \\ y \end{pmatrix}$$ [For $\lambda = 8$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 8x \\ 8y \end{pmatrix}$$

$\Rightarrow 5x - 3y = 8x$

$\quad -3x + 5y = 8y$

$\Rightarrow -3x - 3y = 0$

$\quad -3x - 3y = 0$

$\Rightarrow x + y = 0$

$\quad x + y = 0$

$\Rightarrow x + y = 0$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 $\therefore x = -1$

$\therefore v_1 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$; It is not unit vector or normal vector. Since the length of $\mathbf{v_1}$ is $|\mathbf{v_1}| = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{v_1}$ has to be converted into unit vector by dividing its magnitude or its length.

We have, $\mathbf{v_1} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

$$\therefore \mathbf{v_1} = \begin{pmatrix} \frac{-1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$ [Dividing by its magnitude or its length $\sqrt{2}$]

$$\mathbf{v_1} = \begin{pmatrix} \frac{-\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$ *[Multiplying by $\sqrt{2}$ on numerator and denominator]*

Similarly,

We have, $\mathbf{Av_2 = \lambda v_2}$ $\quad$ $[AV = \lambda V]$

Let $\mathbf{v_2} = \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix}$

$\therefore \mathbf{A^T A v_2 = \lambda v_2}$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \lambda\begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 2\begin{pmatrix} x \\ y \end{pmatrix}$$ [For $\boldsymbol{\lambda = 2}$ & from (iv) putting the value of $A^T A$]

$$\begin{pmatrix} 5 & -3 \\ -3 & 5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\begin{pmatrix} 5x - 3y \\ -3x + 5y \end{pmatrix} = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

$$\Rightarrow 3x - 3y = 0$$
$$-3x + 3y = 0$$

$$\Rightarrow x - y = 0$$
$$x - y = 0$$

$$\Rightarrow x - y = 0$$

In echelon form, there are only one equation in two unknowns, then the system has a non-zero solution and in particular (2-1) =1 free variable, which is y.

Let y = 1 $\therefore \mathbf{x = 1}$

$\therefore v_2 = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ It is not unit vector or normal vector. Since the length of $\mathbf{v_2}$ is $|\mathbf{v_2}| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$; this vector $\mathbf{v_2}$ has to be converted into unit vector by dividing its magnitude or its length.

We have,

$$\mathbf{v_2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Thus, $\mathbf{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ [Dividing by its magnitude or its length $\sqrt{2}$]

$\Rightarrow \mathbf{v}_2 = \begin{pmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$ *[Multiplying by $\sqrt{2}$ on numerator and denominator]*

Now we need to construct the augmented orthogonal matrix V,

$$\mathbf{V} = (\mathbf{v}_1 \quad \mathbf{v}_2) = \begin{pmatrix} \frac{-\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\therefore V^T = \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \quad \text{-------------------------------------(vi)}$$

From (ii), (iii) & (vi), Putting the values of U, S, V^T **in the relation** $\mathbf{A = USV^T}$

Therefore $\mathbf{A = USV^T}$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \textit{Answer}$$

Justification Test:

$$\mathbf{A = USV^T}$$

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \text{[Putting the value of U}, V^T, \text{S]}$$

$$\mathbf{A = USV^T}$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 & 2\sqrt{2} \times \frac{\sqrt{2}}{2} + 0 \\ 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} & 0 + \sqrt{2} \times \frac{\sqrt{2}}{2} \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 1 & 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} -1 \times (-2) + 0 & (-1) \times 2 + 0 \\ 0 + 1 & 0 + 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & -2 \\ 1 & 1 \end{pmatrix}$$ ***[Given Matrix]*** **Proved**

Example 69: Compute Singular Value Decomposition (SVD) for the given matrix

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

In order to find U, In order to find U, Now we need to find the eigenvectors for AA^T
Given,

$$A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}$$

So, $$AA^T = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix} \quad \text{------------------------------(i)}$$

Now we need find the eigenvalues of AA^T

$$|AA^T - \lambda I| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right| = 0$$

$$\left| \begin{pmatrix} 11 & 1 \\ 1 & 11 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

Therefore,

$$\begin{vmatrix} 11-\lambda & 1-0 \\ 1-0 & 11-\lambda \end{vmatrix} = 0$$

$$(11-\lambda)(11-\lambda) - 1 = 0$$

$$\Rightarrow 121 - 11\lambda - 11\lambda + \lambda^2 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 121 - 1 = 0$$

$$\Rightarrow \lambda^2 - 22\lambda + 120 = 0$$

$$\Rightarrow \lambda^2 - 10\lambda - 12\lambda + 120 = 0$$

$$\Rightarrow (\lambda - 10)(\lambda - 12) = 0$$

$$\Rightarrow \lambda = 10,12$$

Therefore eigenvalues are 10 and 12. We construct matrix D by placing eigenvalues along the main diagonal in decreasing order.

$$D = \begin{pmatrix} 12 & 0 \\ 0 & 10 \end{pmatrix}$$

Therefore, taking the square root of matrix D gives,

Since large eigenvalues carry significant representation of the matrix, we arrange the values of the Eigen Values from the largest to the smallest and drop/ignore the values towards the last since small eigenvalues are less indicator.

What is the use of SVD?

01. SVD can be used to compute optimal low-rank approximations of arbitrary matrices.
02. Face recognition: represent the face images as eigenfaces and compute distance between the query face image in the principal component space.

B. LU Decomposition

A procedure for decomposing a $N \times N$ matrix A into a product of a lower triangular matrix L and an upper triangular matrix U,

$$\mathbf{A} = \mathbf{LU}$$

$$\begin{pmatrix} \mathbf{a}_{11} & \mathbf{a}_{12} & \mathbf{a}_{13} \\ \mathbf{a}_{21} & \mathbf{a}_{22} & \mathbf{a}_{23} \\ \mathbf{a}_{31} & \mathbf{a}_{32} & \mathbf{a}_{33} \end{pmatrix} = \begin{pmatrix} \mathbf{l}_{11} & \mathbf{0} & \mathbf{0} \\ \mathbf{l}_{21} & \mathbf{l}_{22} & \mathbf{0} \\ \mathbf{l}_{31} & \mathbf{l}_{32} & \mathbf{l}_{33} \end{pmatrix} \begin{pmatrix} \mathbf{1} & \mathbf{u}_{12} & \mathbf{u}_{13} \\ \mathbf{0} & \mathbf{1} & \mathbf{u}_{23} \\ \mathbf{0} & \mathbf{0} & \mathbf{1} \end{pmatrix}$$

LU decomposition is implemented in *Mathematica* as LU Decomposition$[m]$.

Example 70

Solve the linear system of equations using LU decomposition

$$\boldsymbol{u_1 + 2u_2 + 3u_3 = 5}$$
$$\boldsymbol{2u_1 - 4u_2 + 6u_3 = 18}$$
$$\boldsymbol{3u_1 - 9u_2 - 3u_3 = 6}$$

Solution:

Given the linear system of equations

$$\boldsymbol{u_1 + 2u_2 + 3u_3 = 5}$$
$$\boldsymbol{2u_1 - 4u_2 + 6u_3 = 18}$$
$$\boldsymbol{3u_1 - 9u_2 - 3u_3 = 6}$$

This can be written as

$\mathbf{Au = b}$ --------------------------------(i)

Where,

$$\mathbf{A}=\begin{pmatrix}1 & 2 & 3\\ 2 & -4 & 6\\ 3 & -9 & -3\end{pmatrix}\quad \mathbf{u}=\begin{pmatrix}u_1\\ u_2\\ u_3\end{pmatrix},\quad \mathbf{b}=\begin{pmatrix}5\\ 18\\ 6\end{pmatrix}$$

Again, According to LU decomposition,

$\mathbf{A}=\mathbf{LU}$ ------------------------------(ii)

Where,

$$\mathbf{L}=\begin{pmatrix}l_{11} & 0 & 0\\ l_{21} & l_{22} & 0\\ l_{31} & l_{32} & l_{33}\end{pmatrix}\quad \text{and}\quad \mathbf{U}=\begin{pmatrix}1 & u_{12} & u_{13}\\ 0 & 1 & u_{23}\\ 0 & 0 & 1\end{pmatrix}$$

From (ii), $\mathbf{A}=\mathbf{LU}$

$\Rightarrow \mathbf{LU}=\mathbf{A}$

$$\begin{pmatrix}l_{11} & 0 & 0\\ l_{21} & l_{22} & 0\\ l_{31} & l_{32} & l_{33}\end{pmatrix}\begin{pmatrix}1 & u_{12} & u_{13}\\ 0 & 1 & u_{23}\\ 0 & 0 & 1\end{pmatrix}=\begin{pmatrix}1 & 2 & 3\\ 2 & -4 & 6\\ 3 & -9 & -3\end{pmatrix}$$

$$\begin{pmatrix}l_{11}+0+0 & l_{11}u_{12}+0+0 & l_{11}u_{13}+0+0\\ l_{21}+0+0 & l_{21}u_{12}+l_{22}+0 & l_{21}u_{13}+l_{22}u_{23}+0\\ l_{31}+0+0 & l_{31}u_{12}+l_{32}+0 & l_{31}u_{13}+l_{32}u_{23}+l_{33}\end{pmatrix}=\begin{pmatrix}1 & 2 & 3\\ 2 & -4 & 6\\ 3 & -9 & -3\end{pmatrix}$$

Now,

We can write

$l_{11}=1$ ------------------------------------(iii)

$l_{11}u_{12}=2$ ------------------------------------(iv)

$l_{11}u_{13}=3$ ------------------------------------(v)

$l_{21}=2$ ------------------------------------(vi)

$l_{21}u_{12}+l_{22}=-4$ ------------------------------------(vii)

$l_{21}u_{13}+l_{22}u_{23}=6$ ------------------------------------(viii)

$l_{31}=3$ ------------------------------------(ix)

$l_{31}u_{12}+l_{32}=-9$ ------------------------------------(x)

$l_{31}u_{13}+l_{32}u_{23}+l_{33}=-3$ ------------------------------------(xi)

From (iii) & (iv)

$$l_{11} = 1$$

$$l_{11}u_{12} = 2$$

$$1.u_{12} = 2$$

$u_{12} = 2$ ------------------------------------(xii)

From (iii) & (v)

$$l_{11}u_{13} = 3$$

$$1.u_{13} = 3$$

$u_{13} = 3$ ------------------------------------(xiii)

From (vii),

$$l_{21}u_{12} + l_{22} = -4$$

$2.2 + l_{22} = -4$ $[u_{12} = 2 ; l_{21} = 2]$

$$4 + l_{22} = -4$$

$\therefore l_{22} = -8$ ------------------------------------(xiv)

From (viii),

$$l_{21}u_{13} + l_{22}u_{23} = 6$$

$2.3 - 8u_{23} = 6$ $[l_{21} = 2 ; u_{13} = 3 ; \therefore l_{22} = -8]$

$$-8u_{23} = 6 - 6$$

$u_{23} = 0$ ------------------------------------(xv)

From (x),

$$l_{31}u_{12} + l_{32} = -9$$

$3.2 + l_{32} = -9$ $[l_{31} = 3 ; u_{12} = 2]$

$$l_{32} = -9 - 6$$

$l_{32} = -15$ ------------------------------------(xvi)

From (xi)

$$l_{31}u_{13} + l_{32}u_{23} + l_{33} = -3$$

$3.3 - 15.0 + l_{33} = -3$ $[l_{31} = 3 ; u_{13} = 3 ; l_{32} = -15 ; u_{23} = 0]$

$$l_{33} = -3 - 9$$

$\mathbf{l_{33} = -12}$ --------------------------------------(xvii)

$$\therefore \mathbf{L} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix}$$

$$\Rightarrow \mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix}$$

And

$$\mathbf{U} = \begin{pmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

$$\mathbf{U} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

From (i),

$\mathbf{Au = b}$

$\mathbf{LUu = b}$ [from (ii), A = LU]

$\mathbf{LUu = b}$ --------------------------------------(xviii)

Let,

$\mathbf{Uu = v}$ --------------------------------------(xix)

Where,

Where, $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$

From (xviii),

$\mathbf{LUu = b}$
$\mathbf{Lv = b}$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & -8 & 0 \\ 3 & -15 & -12 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$\Rightarrow\begin{pmatrix} 1\times v_1+0\times v_2+0\times v_3 \\ 2\times v_1-8\times v_2+0\times v_3 \\ 3\times v_1-15\times v_2-12\times v_3 \end{pmatrix}=\begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$$\Rightarrow\begin{pmatrix} v_1 \\ 2v_1-8v_2 \\ 3v_1-15v_2-12v_3 \end{pmatrix}=\begin{pmatrix} 5 \\ 18 \\ 6 \end{pmatrix}$$

$v_1 = 5$ --------------------------------------(xx)

$2v_1 - 8v_2 = 18$ --------------------------------------(xxi)

$3v_1 - 15v_2 - 12v_3 = 6$ --------------------------------------(xxii)

From (xxi),

$2v_1 - 8v_2 = 18$

$2.5 - 8v_2 = 18$ $[\, v_1 = 5 \,]$

$10 - 8v_2 = 18$

$-8v_2 = 18 - 10$

$-8v_2 = 8$

$\therefore v_2 = -1$ --------------------------------------(xxiii)

From (xxii),

$3v_1 - 15v_2 - 12v_3 = 6$

$3.5 - 15(-1) - 12v_3 = 6$ $[\, v_1 = 5\,;\, v_2 = -1 \,]$

$15 + 15 - 12v_3 = 6$

$30 - 12v_3 = 6$

$-12v_3 = 6 - 30$

$-12v_3 = -24$

$\therefore v_3 = 2$ --------------------------------------(xxiv)

From (xix),

$Uu = v$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1\times u_1 + 2\times u_2 + 3\times u_3 \\ 0\times u_1 + 1\times u_2 + 0\times u_3 \\ 0\times u_1 + 0\times u_2 + 1\times u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} u_1 + 2u_2 + 3u_3 \\ u_2 \\ u_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$$

$\mathbf{u_1 + 2u_2 + 3u_3 = 5}$ -------------------------------------(xxv)

$\mathbf{u_2 = -1}$ -------------------------------------(xxvi)

$\mathbf{u_3 = 2}$ -------------------------------------(xxvii)

From (xxv),

$\mathbf{u_1 + 2u_2 + 3u_3 = 5}$

$\mathbf{u_1 + 2(-1) + 3.2 = 5}$

$\mathbf{u_1 - 2 + 6 = 5}$

$\mathbf{u_1 = 5 - 4}$

$\mathbf{u_1 = 1}$

$\mathbf{u_1 = 1}$, $\mathbf{u_2 = -1}$, $\mathbf{u_3 = 2}$ Answer

Example 71: Given the following system of linear equations, determine the value of each of the variables using the LU decomposition method.

$$6x_1 - 2x_2 = 14$$
$$9x_1 - x_2 + x_3 = 21$$
$$3x_1 - 7x_2 + 5x_3 = 9$$

Hence, $\vec{A}.(\vec{B}\times\vec{C}) = -\vec{A}.(\vec{C}\times\vec{B})$

That is, $[ABC] = -[ACB]$

Q # 12: $\vec{B} = B_x\hat{i}+B_y\hat{j}+B_z\hat{k}$, **If** $\vec{A} = A_x\hat{i}+A_y\hat{j}+A_z\hat{k}$ and Find $\vec{A}\times\vec{B}$

Answer: :

$$\vec{A}\times\vec{B} = (A_x\hat{i}+A_y\hat{j}+A_z\hat{k})\times(B_x\hat{i}+B_y\hat{j}+B_z\hat{k})$$

$$= A_xB_x(\hat{i}\times\hat{i}) + A_xB_y(\hat{i}\times\hat{j}) + A_xB_z(\hat{i}\times\hat{k}) + A_yB_x(\hat{j}\times\hat{i}) + A_yB_y(\hat{j}\times\hat{j}) + A_yB_z(\hat{j}\times\hat{k})$$
$$+ A_zB_x(\hat{k}\times\hat{i}) + A_zB_y(\hat{k}\times\hat{j}) + A_zB_z(\hat{k}\times\hat{k})$$

$$= A_xB_x\times 0 + A_xB_y(\hat{k}) + A_xB_z(-\hat{j}) + A_yB_x(-\hat{k}) + A_yB_y\times 0 + A_yB_z(\hat{i})$$
$$+ A_zB_x(\hat{j}) + A_zB_y(-\hat{i}) + A_zB_z\times 0$$

$$= A_xB_y\hat{k} - A_xB_z\hat{j} - A_yB_x\hat{k} + A_yB_z\hat{i} + A_zB_x\hat{j} - A_zB_y\hat{i}$$

$$= A_xB_y\hat{k} - A_xB_z\hat{j} - A_yB_x\hat{k} + A_yB_z\hat{i} + A_zB_x\hat{j} - A_zB_y\hat{i}$$

$$= A_yB_z\hat{i} - A_zB_y\hat{i} - A_xB_z\hat{j} + A_zB_x\hat{j} + A_xB_y\hat{k} - A_yB_x\hat{k}$$

$$\vec{A}\times\vec{B} = \hat{i}(A_yB_z - A_zB_y) - \hat{j}(A_xB_z - A_zB_x) + \hat{k}(A_xB_y - A_yB_x) \text{------------------------(i)}$$

$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \text{--(ii)}$$

Q # 13: Find a unit vector perpendicular to the vectors $\vec{a} = 3\hat{i}+\hat{j}$ and $\vec{b} = -\hat{i}+2\hat{j}+2\hat{k}$

Answer: A vector perpendicular to $\vec{a}$ and $\vec{b}$ is $\vec{a}\times\vec{b}$

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 0 \\ -1 & 2 & 2 \end{vmatrix} = \hat{i}(2-0) - \hat{j}(6-0) + \hat{k}(6+1) = 2\hat{i} - 6\hat{j} + 7\hat{k}$$

A unit vector, perpendicular to $\vec{a}$ and $\vec{b}$, in this direction is obtained by simply dividing $\vec{a}\times\vec{b}$ by its magnitude. Thus

$$\hat{e} = \frac{2\hat{i}-6\hat{j}+7\hat{k}}{\sqrt{2^2+(-6)^2+7^2}} = \frac{2\hat{i}-6\hat{j}+7\hat{k}}{\sqrt{89}}$$ **is the required vector.**

$\therefore$ Magnitude of Unit vector of $\hat{e}$

$$= \left|\hat{e}\right| = \sqrt{\left(\frac{2}{\sqrt{89}}\right)^2 + \left(\frac{-6}{\sqrt{89}}\right)^2 + \left(\frac{7}{\sqrt{89}}\right)^2} = \sqrt{\frac{4}{89} + \frac{36}{89} + \frac{49}{89}} = \sqrt{\frac{89}{89}} = 1$$

Home Task

Find a unit vector parallel to the resultant of vectors, $\vec{A} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{B} = \hat{i} + 2\hat{j} + 3\hat{k}$

$$\hat{e} = \frac{A + B}{|A + B|}$$

Q#14: Show that $\vec{\mathbf{A}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - 3\hat{\mathbf{k}}$, $\vec{\mathbf{B}} = 2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$ and $\vec{\mathbf{C}} = 3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}$ are coplanar.

Answer:

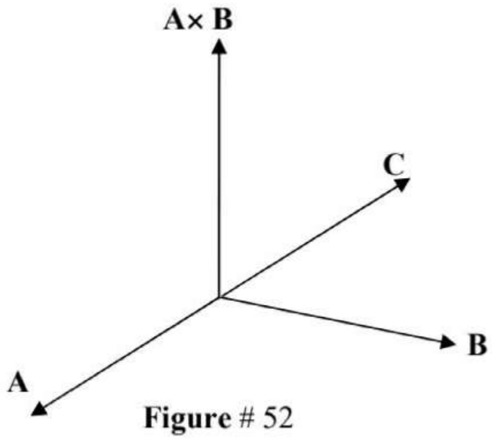

Figure # 52

If **A** is a third vector perpendicular to **(B×C),** then **A, B** and **C** are coplanar and **A. (B×C) =0**

Therefore, three vectors **A, B, C** are coplanar if **A. (B×C) = 0**

$$\mathbf{B\times C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 2 \\ 3 & 1 & -1 \end{vmatrix} == \hat{i}(1-2) - \hat{j}(-2-6) + \hat{k}(2+3) = -\hat{i} + 8\hat{j} + 5\hat{k}$$

$$\mathbf{A.\ (B\times C)} = (\hat{i} + 2\hat{j} - 3\hat{k}).(-\hat{i} + 8\hat{j} + 5\hat{k}) = -1+16-15=0$$

Therefore A, B, C are coplanar.

Q#15: On which condition of λ the vectors $\hat{i}+2\hat{j}+3\hat{k}$, $\lambda\hat{i}+4\hat{j}+7\hat{k}$, $-3\hat{i}-2\hat{j}-5\hat{k}$ are collinear

Solution

We know that if the points $(x_1, y_1, z_1), (x_2, y_2, z_2)$ and (x_3, y_3, z_3) be collinear then

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 2 & 3 \\ \lambda & 4 & 7 \\ -3 & -2 & -5 \end{vmatrix} = 0$$

$$\Rightarrow 1[4\times(-5)-7\times(-2)]-2[\lambda(-5)-(-3)\times 7]+3[\lambda(-2)-4(-3)]=0$$
$$\Rightarrow 1[-20+14]-2[-5\lambda+21]+3[-2\lambda+12]=0$$
$$\Rightarrow -6+10\lambda-42-6\lambda+36=0$$
$$\Rightarrow 4\lambda-48+36=0$$
$$\Rightarrow 4\lambda-12=0$$
$$\Rightarrow 4\lambda=12$$
$$\Rightarrow \lambda=3$$

Q# 16: Figure no 53 shows a force $\vec{\mathbf{F}}$ of 100 N applied in the positive z-direction at the point **Q(1,1,1)** of a cube whose sides have a length of 1 m. assuming that the cube is free to rotate about the point **P(0,0,0)** (the origin), find the scalar moment of the force about **P** and describe the direction of rotation.

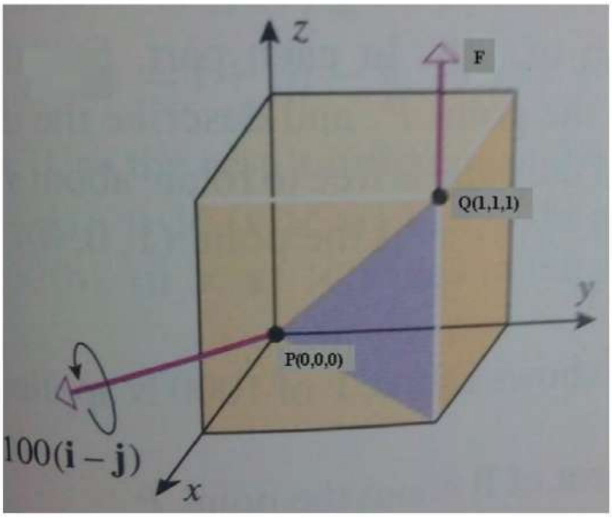

Figure 53

Answer: The force vector $\vec{F} = 0.\hat{i} + 0.\hat{j} + 100\hat{k}$ and the vector from P to Q is

$\vec{PQ} = \hat{i} + \hat{j} + \hat{k}$, so the vector moment of $\vec{F}$ about P is

$$\vec{PQ} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & 0 & 100 \end{vmatrix} = \hat{i}(100-0) - \hat{j}(100-0) + \hat{k}(0-0)$$

$$= 100\hat{i} - 100\hat{j}$$

Thus the scalar moment of $\vec{F}$ about P is

$$\left|\vec{PQ} \times \vec{F}\right| = \sqrt{(100)^2 + (-100)^2} = \sqrt{10000 + 10000} = \sqrt{20000} = \sqrt{2 \times (100)^2} = 100\sqrt{2} \text{ N.m}$$

and the direction of rotation is counterclockwise looking along the vector $100\hat{i} - 100\hat{j}$ $= 100(\hat{i} - \hat{j})$ towards its initial point **(Figure no 53)**

Q-17: If $\vec{a}.\vec{b} = \sqrt{3}$ and $\vec{a} \times \vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$, find the angle between $\vec{a}$ and $\vec{b}$

Answer: We have, $\vec{a}.\vec{b} = |\vec{a}||\vec{b}|\cos\theta$

Given, $\vec{a}.\vec{b} = \sqrt{3}$

$\therefore |\vec{a}||\vec{b}|\cos\theta = \sqrt{3}$ ------------------------------------(i)

Again, we have, $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{\eta}$

Given, $\vec{a} \times \vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$

$|\vec{a}||\vec{b}|\sin\theta\,\hat{\eta} = \hat{i} + 2\hat{j} + 2\hat{k}$

$\hat{\eta}|\vec{a}||\vec{b}|\sin\theta = \hat{i} + 2\hat{j} + 2\hat{k}$ --------------------------------------(ii)

We have, $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{\eta}$

Again, we can write, $\hat{\eta} = \dfrac{\vec{a} \times \vec{b}}{|\vec{a}||\vec{b}|\sin\theta} = \dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$ -------------------------------------**(iii)**

Given, $\vec{a} \times \vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$

$\therefore \left|\vec{a}\times\vec{b}\right| = \sqrt{1^2+2^2+2^2} = \sqrt{9} = 3$

From (iii),

$$\hat{\eta} = \frac{\vec{a}\times\vec{b}}{\left|\vec{a}\times\vec{b}\right|}$$

$$\hat{\eta} = \frac{\hat{i}+2\hat{j}+2\hat{k}}{3}$$

Putting the value of $\hat{\eta}$ **in (ii), we get,**

$$\hat{\eta}\left|\vec{a}\right|\left|\vec{b}\right|\sin\theta = \hat{i}+2\hat{j}+2\hat{k}$$

$$\frac{\hat{i}+2\hat{j}+2\hat{k}}{3}\times\left|\vec{a}\right|\left|\vec{b}\right|\sin\theta = \hat{i}+2\hat{j}+2\hat{k}$$

$$\frac{1}{3}\times\left|\vec{a}\right|\left|\vec{b}\right|\sin\theta = 1$$

$$\left|\vec{a}\right|\left|\vec{b}\right|\sin\theta = 3 \qquad \textbf{----------------------------------(iv)}$$

(iv) $\div$ **(i)**

$$\frac{\left|\vec{a}\right|\left|\vec{b}\right|\sin\theta}{\left|\vec{a}\right|\left|\vec{b}\right|\cos\theta} = \frac{3}{\sqrt{3}}$$

$$\frac{\sin\theta}{\cos\theta} = \sqrt{3}$$

$$\tan\theta = \sqrt{3}$$

$$\tan\theta = \tan 60^0$$

$\therefore \theta = 60^0$ **Answer**

Q-18: Find all vectors $\vec{v}$ **such that** $(\hat{i}+2\hat{j}+\hat{k})\times\vec{v} = 3\hat{i}+\hat{j}-5\hat{k}$

Answer:

Given,

$$(\hat{i}+2\hat{j}+\hat{k})\times\vec{v} = 3\hat{i}+\hat{j}-5\hat{k} \text{-----------------(i)}$$

Let $\vec{v} = x\hat{i}+y\hat{j}+z\hat{k}$

$$(\hat{i}+2\hat{j}+\hat{k})\times\vec{v}=(\hat{i}+2\hat{j}+\hat{k})\times x\hat{i}+y\hat{j}+z\hat{k}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & 1\\ x & y & z\end{vmatrix}$$

$$\therefore (\hat{i}+2\hat{j}+\hat{k})\times\vec{v}=\hat{i}(2z-y)-\hat{j}(z-x)+\hat{k}(y-2x) \text{ ---------------------- (ii)}$$

Given,

$(\hat{i}+2\hat{j}+\hat{k})\times\vec{v}=3\hat{i}+\hat{j}-5\hat{k}$

From (ii), We can write,

$\therefore (\hat{i}+2\hat{j}+\hat{k})\times\vec{v}=\hat{i}(2z-y)-\hat{j}(z-x)+\hat{k}(y-2x)=3\hat{i}+\hat{j}-5\hat{k}$

$\Rightarrow \hat{i}(2z-y)-\hat{j}(z-x)+\hat{k}(y-2x)=3\hat{i}+\hat{j}-5\hat{k}$

Equating the coefficient of $\hat{i}$, $\hat{j}$, and $\hat{k}$ on both sides

$\therefore 2z-y=3$

$-(z-x)=1$

$y-2x=-5$

That is,

$$\begin{aligned} &\therefore 2z-y=3\\ \Rightarrow\ &x-z=1\\ &y-2x=-5\end{aligned}$$

$$\begin{aligned} &x-z=1\\ \Rightarrow\ &y-2x=-5\\ &2z-y=3\end{aligned}$$

$$\begin{aligned} &x+0.y-z=1\\ \Rightarrow\ &-2x+y+0.z=-5 \text{ ------------------------------------- (iii)}\\ &0.x-y+2z=3\end{aligned}$$

We have,

$\mathbf{L_i \rightarrow -a_{i1}L_1 + a_{11}L_i}$

Here, $a_{11}=1,\ a_{12}=0,\ a_{13}=-1,\ a_{21}=-2,\ a_{22}=1,\ a_{23}=0,\ a_{31}=0,\ a_{32}=-1,\ a_{33}=2$

1st time:

$\mathbf{i=2,\ L_2 \rightarrow -a_{21}L_1 + a_{11}L_2}$

$=\ -(-2)(x+0.y-z-1)+1(-2x+y+0.z+5)$

$=2x-2z-2-2x+y+5$

$=y-2z+3$

$\mathbf{i = 3,\ L_3 \rightarrow -a_{31}L_1 + a_{11}L_3}$

$= \ -0(x + 0.y - z - 1) + 1(0.x - y + 2z - 3)$

$= -y + 2z - 3$

Thus we obtain the following new system

$$x + 0.y - z = 1$$
$$y - 2z = -3$$
$$-y + 2z = 3$$

2^nd^ time:

$x \ + \ 0.y - z = 1$

$y - 2z = -3$ ------------- L_1

$-y + 2z = 3$ ------------- L_2

$$\overset{\overset{a_{11}}{\uparrow}}{1}y - \overset{\overset{a_{12}}{\uparrow}}{2}z = -3$$
$$-\overset{\overset{a_{21}}{\uparrow}}{1}y + \overset{\overset{a_{22}}{\uparrow}}{2}z = 3$$

We have,

$\mathbf{L_i \rightarrow -a_{i1}L_1 + a_{11}L_i}$

Here, $a_{11} = 1,\ a_{12} = -2,\ a_{21} = -1,\ a_{22} = 2$

$\mathbf{i = 2,\ L_2 \rightarrow -a_{21}L_1 + a_{11}L_2}$

$= \ -(-1)(y - 2z + 3) + 1(-y + 2z - 3)$

$= y - 2z + 3 - y + 2z - 3$

$= 0$

Thus we obtain the following new system

$$x + 0.y - z = 1$$
$$y - 2z = -3$$
$$0 = 0$$

Thus we obtain the following new system

$$x + 0.y - z = 1$$
$$y - 2z = -3$$

In echelon form, there are only two equations in three unknowns, then the system has a non-zero solution and in particular (3-2)=1 free variable, which is z. We obtain more than one solution of the system; hence we will get infinite number of vectors $\vec{v}$

$$\Delta\vec{r} = \vec{r}(t_2) - \vec{r}(t_1) \qquad \text{--------------------------(i)}$$

The displacement vector, which describes the change in position of the particle during the time interval, can be obtained by integrating the velocity function from $\mathbf{t_1}$ to $\mathbf{t_2}$.

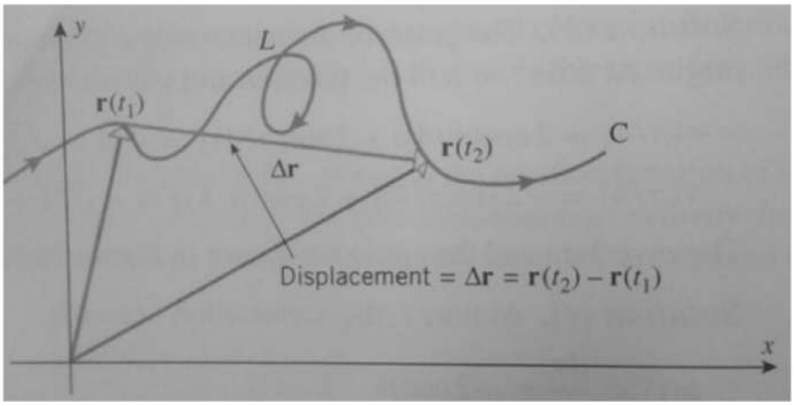

Figure 58

$$\Delta\vec{r} = \int_{t_1}^{t_2} \vec{v}(t)\,dt = \int_{t_1}^{t_2} \frac{dr}{dt}\,dt = \left[\vec{r}(t)\right]_{t_1}^{t_2} = \vec{r}(t_2) - \vec{r}(t_1) \text{ --------------(ii)}$$

The distance travelled by the particle over the time interval $\mathbf{t_1 \le t \le t_2}$ is:

$$s = \int_{t_1}^{t_2} \left|\frac{dr}{dt}\right| dt = \int_{t_1}^{t_2} \left|\vec{v}(t)\right| dt \qquad \text{-------------------------(iii)}$$

Q#19: A particle moves along a curve whose parametric equations are $\mathbf{x = e^{-t}, y = 2\cos 3t, z = 2\sin 3t,}$ Where t is the time.

(a) Determine its velocity and acceleration at any time
(b) Find the magnitudes of the velocity and acceleration at t = 0.

Answer: The position vector $\vec{r}$ of the particle is $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ [Page no 48, Figure no 57, Equation no (i)]

$$\Rightarrow \vec{r} = e^{-t}\,\hat{i} + 2\cos 3t\,\hat{j} + 2\sin 3t\,\hat{k}$$

Then the velocity is $\vec{V} = \dfrac{d\vec{r}}{dt} = \dfrac{d}{dt}(e^{-t})\,\hat{i} + \dfrac{d}{dt}(2\cos 3t)\,\hat{j} + \dfrac{d}{dt}(2\sin 3t)\,\hat{k}$

$$\frac{d\vec{r}}{dt} = -e^{-t}\,\hat{i} - 6\sin 3t\,\hat{j} + 6\cos 3t\,\hat{k} \qquad [\frac{d}{dx}(e^{mx}) = me^{mx}]$$

and the acceleration is: $\vec{a} = \dfrac{d\vec{V}}{dt} = \dfrac{d}{dt}(\dfrac{d\vec{r}}{dt}) = \dfrac{d}{dt}(-e^{-t}\,\hat{i} - 6\sin 3t\,\hat{j} + 6\cos 3t\,\hat{k})$

$$\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2} = e^{-t}\hat{i} - 18\cos 3t\,\hat{j} - 18\sin 3t\,\hat{k}$$

b) At $t = 0$,

Then the velocity is $\quad \vec{V} = \frac{d\vec{r}}{dt} = -e^{-t}\hat{i} - 6\sin 3t\,\hat{j} + 6\cos 3t\,\hat{k}$

$$\vec{V} = \frac{d\vec{r}}{dt} = -e^{-0}\hat{i} - 6\sin 3\times 0\,\hat{j} + 6\cos 3\times 0\,\hat{k} \qquad [t = 0]$$

$$\vec{V} = \frac{d\vec{r}}{dt} = -e^{-0}\hat{i} - 6\sin 0 + 6\cos 0\,\hat{k}$$

$$\vec{V} = \frac{d\vec{r}}{dt} = -\hat{i} + 6\hat{k} \quad [e^{-0} = \frac{1}{e^0} = 1;\ \sin 0 = 0;\ \cos 0 = 1]$$

and the acceleration is:

$$\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2} = e^{-t}\hat{i} - 18\cos 3t\,\hat{j} - 18\sin 3t\,\hat{k}$$

$$\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2} = e^{-0}\hat{i} - 18\cos 3\times 0\,\hat{j} - 18\sin 3\times 0\,\hat{k} \qquad [t = 0]$$

$$\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{r}}{dt^2} = \hat{i} - 18\hat{j} \quad [e^{-0} = \frac{1}{e^0} = 1;\ \sin 0 = 0;\ \cos 0 = 1]$$

The Magnitude of the velocity is $\quad \left|\vec{V}\right| = \sqrt{(-1)^2 + (6)^2} = \sqrt{37}$

The Magnitude of the acceleration is $\left|\vec{a}\right|$: $\sqrt{(1)^2 + (-18)^2} = \sqrt{325}$ Answer

Q#20: A particle moves along a curve whose parametric equations are $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is time. Find the component of the velocity at time t = 1 in the direction $\vec{a} = \hat{i} - 3\hat{j} + 2\hat{k}$

Answer:

The position vector $\vec{r}$ of the particle is $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ [Page no 48, Figure no 57, Equation no (i)]

$$\Rightarrow \vec{r} = 2t^2\,\hat{i} + (t^2 - 4t)\,\hat{j} + (3t - 5)\,\hat{k}$$

Then the velocity is $\quad \vec{V} = \frac{d\vec{r}}{dt} = \frac{d}{dt}[2t^2\,\hat{i} + (t^2 - 4t)\,\hat{j} + (3t - 5)\,\hat{k}]$

$$\vec{a} = -\omega^2 \vec{r} \text{ --(i)}$$

$$\vec{a} \propto \vec{r}$$

$$|\vec{a}| \propto |\vec{r}| \text{ ---(ii)}$$

From equation (1), the acceleration is opposite to the direction of $\vec{r}$. i.e. it is directed toward the origin, Its magnitude is proportional to $|\vec{r}|$ which is the distance from the origin.

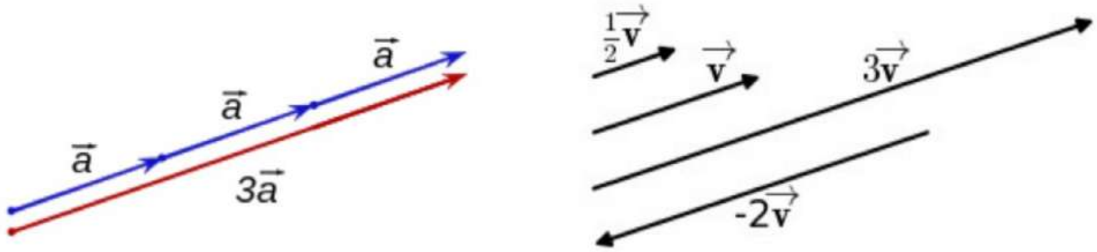

Figure # 60

c) Here $\vec{r} = \cos\omega t\,\hat{i} + \sin\omega t\,\hat{j}$ and $\vec{V} = -\omega\sin\omega t\,\hat{i} + \omega\cos\omega t\,\hat{j}$

$$\vec{r} \times \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\omega t & \sin\omega t & 0 \\ -\omega\sin\omega t & \omega\cos\omega t & 0 \end{vmatrix}$$

$$= \hat{i}(\sin\omega t \times 0 - 0 \times \omega\cos\omega t) - \hat{j}(\cos\omega t \times 0 - (-\omega\sin\omega t) \times 0 + \hat{k}(\cos\omega t \times \omega\cos\omega t$$
$$-(-\omega\sin\omega t) \times \sin\omega t)$$

$$= \hat{i} \times 0 - \hat{j} \times 0 + \hat{k}(\omega\cos^2\omega t + \omega\sin^2\omega t)$$

$$= \hat{k}\,\omega(\cos^2\omega t + \sin^2\omega t)$$

$$= \hat{k}\,\omega.1 \qquad [\because \cos^2\omega t + \sin^2\omega t = 1]$$

$$= \hat{k}\,\omega$$

Q#22: A particle moves along a circular path in such a way that its x- and y-coordinates at time t are $x = 2\cos t$, $y = 2\sin t$

a) Find the instantaneous velocity and speed of the particle at time t.

b) Sketch the path of the particle and show the position and velocity vectors at time $t = \frac{\pi}{4}$ with the velocity vector drawn so that its initial point is at the top of the position vector

c) Show that at each instant the acceleration vector is perpendicular to the velocity vector

Answer:

a)

Let the position vector at any time t is:

$\vec{OP} = \vec{r}(t) = x\hat{i} + y\hat{j}$ **[Page no 48, Figure no 57, Equation no (i)]**

At time t, the position vector is:

$\vec{OP} = \vec{r}(t) = 2\cos t\,\hat{i} + 2\sin t\,\hat{j}$ [Given, $x = 2\cos t$, $y = 2\sin t$] --------------------(i)

So the instantaneous velocity is:

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2\sin t\,\hat{i} + 2\cos t\,\hat{j} \text{ ----------------------------------(ii)}$$

So the instantaneous speed is:

$$\left|\vec{V}(t)\right| = \sqrt{(-2\sin t)^2 + (2\cos t)^2} = \sqrt{4\sin^2 t + 4\cos^2 t} = 2$$

Answer (b)

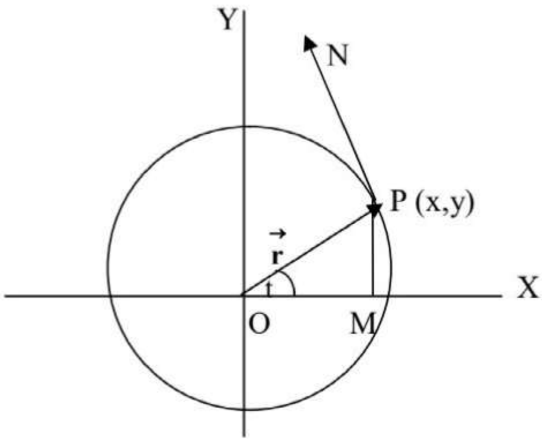

Figure #: 61

$\vec{OP} = \vec{r}(t) = 2\cos t\,\hat{i} + 2\sin t\,\hat{j}$

Here, $\mathbf{OM = x = 2\cos t}$ and $\mathbf{PM = y = 2\sin t}$

$\angle \mathbf{POM = t}$

$$\frac{PM}{OP} = \sin t$$

$$\frac{2\sin t}{OP} = \sin t \text{ [Given } PM = y = 2\sin t \text{]}$$

$$OP \sin t = 2\sin t$$

$$OP = 2 \text{ ---------------------------------(iii)}$$

Similarly,

$$\frac{OM}{OP} = \cos t$$

$$\frac{2\cos t}{OP} = \cos t \qquad [OM = x = 2\cos t]$$

$$OP \cos t = 2\cos t$$

$$OP = 2 \text{ ----------------------------------(iv)}$$

Hence the radius of the circle is OP = 2.

At time $t = \frac{\pi}{4}$, the position and velocity vector of the particles are:

$$\vec{OP} = \vec{r}(t) = 2\cos t\,\hat{i} + 2\sin t\,\hat{j} \qquad \text{[From (i)]}$$

$$\vec{r}(\frac{\pi}{4}) = 2\cos\frac{\pi}{4}\hat{i} + 2\sin\frac{\pi}{4}\hat{j}$$

$$\vec{r}(\frac{\pi}{4}) = 2\frac{1}{\sqrt{2}}\hat{i} + 2\frac{1}{\sqrt{2}}\hat{j}$$

$$\vec{OP} = \vec{r}(\frac{\pi}{4}) = \sqrt{2}\,\hat{i} + \sqrt{2}\,\hat{j} \text{ When } t = \frac{\pi}{4}.$$

From (ii),

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2\sin t\,\hat{i} + 2\cos t\,\hat{j}$$

$$\vec{V}(\frac{\pi}{4}) = -2\sin\frac{\pi}{4}\hat{i} + 2\cos\frac{\pi}{4}\hat{j}$$

$$\vec{PN} = \vec{V}(\frac{\pi}{4}) = -2\frac{1}{\sqrt{2}}\hat{i} + 2\frac{1}{\sqrt{2}}\hat{j} = \vec{V}(\frac{\pi}{4}) = -\sqrt{2}\,\hat{i} + \sqrt{2}\,\hat{j}$$

Answer (c):

We have,

$$\vec{V}(t) = \frac{d\vec{r}}{dt} = -2\sin t\,\hat{i} + 2\cos t\,\hat{j} \qquad \text{[From (ii)]}$$

At time t, the acceleration vector is:

$$\therefore \vec{a}(t) = \frac{d\vec{v}}{dt} = -2\cos t\,\hat{i} - 2\sin t\,\hat{j} \text{ --------------------------(v)}$$

Test: From (ii) & (v),

$\vec{V}(t).\vec{a}(t) = (-2\sin t\,\hat{i} + 2\cos t\,\hat{j}).\,(-2\cos t\,\hat{i} - 2\sin t\,\hat{j}) = 4\sin t\cos t - 4\sin t\cos t = 0$

Since the dot product of Velocity Vector (ii) and acceleration Vector (v) is Zero, Hence acceleration vector is perpendicular to the velocity vector.

(Proved)

Q#23: A particle moves through 3-space in such a way that its velocity is $\vec{v}(t) = \hat{i} + t\,\hat{j} + t^2\,\hat{k}$. Find the co-ordinates of the particle at time $t = 1$ given that the particle is at the point $(-1,2,4)$ at time $t = 0$

Answer: We have,

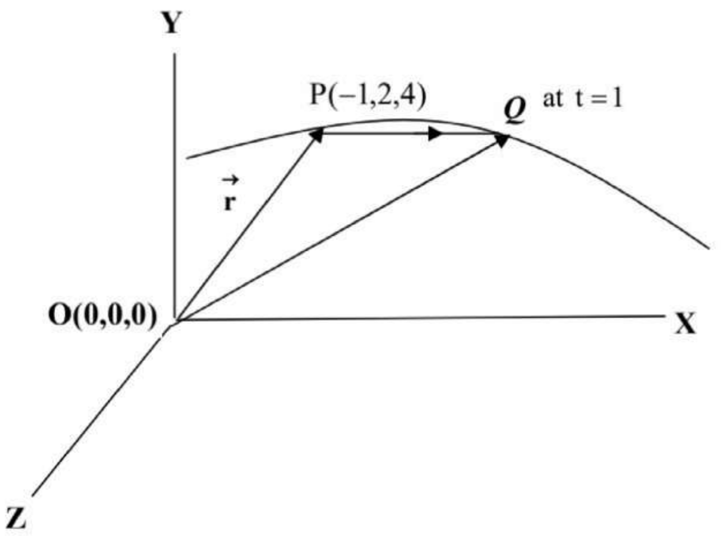

Figure 62

$\vec{v}(t) = \dfrac{d\vec{r}}{dt}$ Where $\vec{r}$ is a position vector.

Given, $\vec{v}(t) = \hat{i} + t\,\hat{j} + t^2\,\hat{k}$

$\therefore\ \vec{v}(t) = \dfrac{d\vec{r}(t)}{dt} = \hat{i} + t\,\hat{j} + t^2\,\hat{k}$

$\dfrac{d\vec{r}(t)}{dt} = \hat{i} + t\,\hat{j} + t^2\,\hat{k}$

$d\vec{r}(t) = (\hat{i} + t\,\hat{j} + t^2\,\hat{k})dt$ -----------------------------(i)

Integrate (i) both sides, we get,

$\int d\vec{r}(t) = \int(\hat{i} + t\,\hat{j} + t^2\,\hat{k})dt$

$\int d\vec{r}(t) = \int\hat{i}\,dt + \int t\,\hat{j}\,dt + \int t^2\,\hat{k}\,dt$

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} + C \quad \text{------------------------(ii)}$$

Where C is a vector constant of integration. Since the coordinates of the particle at time $t = 0$ are $(-1,2,4)$, the position vector at time $t = 0$ is

We have the position vector

$$\vec{r}(t) = x\hat{i} + y\hat{j} + z\hat{k} \qquad \textbf{[Page no 48, Figure no 57, Equation no (i)]}$$

$$\vec{r}(0) = (-1)\hat{i} + 2\hat{j} + 4\hat{k} \qquad \text{[at time } \mathbf{t = 0}\text{, the position vector is at } \mathbf{(-1,2,4)}\text{]}$$

$$\vec{r}(0) = (-1)\hat{i} + 2\hat{j} + 4\hat{k} \quad \text{------------------------(iii)}$$

Again, putting $t = 0$ in (ii), we get,

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} + C$$

$$\vec{r}(0) = 0.\hat{i} + \frac{0^2}{2}\hat{j} + \frac{0^3}{3}\hat{k} + C$$

$$\vec{r}(0) = 0 + 0 + 0 + C$$

$$\vec{r}(0) = C \quad \text{------------------------(iv)}$$

Comparing (iii) and (iv), we get,

$$\vec{r}(0) = (-1)\hat{i} + 2\hat{j} + 4\hat{k} = C$$

$$C = -\hat{i} + 2\hat{j} + 4\hat{k} \quad \text{-------------------------(v)}$$

Putting the value of C in (ii), we get,

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} + C$$

$$\vec{r}(t) = t\hat{i} + \frac{t^2}{2}\hat{j} + \frac{t^3}{3}\hat{k} - \hat{i} + 2\hat{j} + 4\hat{k}$$

$$\vec{r}(t) = (t-1)\hat{i} + (\frac{t^2}{2} + 2)\hat{j} + (\frac{t^3}{3} + 4)\hat{k} \quad \text{------------------(vi)}$$

Thus, at time $t = 1$, the position vector of the particle is

From (vi),

$$\vec{r}(t) = (t-1)\hat{i} + (\frac{t^2}{2} + 2)\hat{j} + (\frac{t^3}{3} + 4)\hat{k}$$

$$\vec{r}(1) = (1-1)\hat{i} + (\frac{1^2}{2} + 2)\hat{j} + (\frac{1^3}{3} + 4)\hat{k}$$

$$\vec{r}(1) = 0\hat{i} + \frac{5}{2}\hat{j} + \frac{13}{3}\hat{k}.$$

So, the coordinates of the particle at time $t = 1$ is $(0, \frac{5}{2}, \frac{13}{3})$ the Answer

Q# 25: If $\phi(x,y,z)=3x^2y-y^3z^2$, find $\vec{\nabla}\Phi$ (or grad Φ) at the point (1,-2,-1).

Answer: $\vec{\nabla}\Phi=(\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})(3x^2y-y^3z^2)$

$$=\hat{i}\frac{\delta}{\delta x}(3x^2y-y^3z^2)+\hat{j}\frac{\delta}{\delta y}(3x^2y-y^3z^2)+\hat{k}\frac{\delta}{\delta z}(3x^2y-y^3z^2)$$

$$=\hat{i}(6xy-0)+\hat{j}\,(3x^2-3y^2z^2)+\hat{k}\,(0-2y^3z)$$

$$=6xy\,\hat{i}+(3x^2-3y^2z^2)\,\hat{j}-2y^3z\,\hat{k}$$

$$=6(1)(-2)\,\hat{i}+\{3(1)^2-3(-2)^2(-1)^2\}\,\hat{j}-2(-2)^3(-1)\,\hat{k}$$

$=-12\hat{i}-9\hat{j}-16\hat{k}$ (Answer).

Q# 26: Find $\vec{\nabla}\phi$ if (a) $\phi=\ln|\vec{r}|$ (b) $\phi=\frac{1}{|\vec{r}|}$

Let $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ **[Page no 48, Figure no 57, Equation no (i)]**

(a) Then $|\vec{r}|=\sqrt{x^2+y^2+z^2}$ and $|\vec{r}|^2=x^2+y^2+z^2$

Then $\phi=\ln|\vec{r}|=\ln\sqrt{x^2+y^2+z^2}=\ln(x^2+y^2+z^2)^{1/2}=\frac{1}{2}\ln(x^2+y^2+z^2)$

$$\therefore\vec{\nabla}\phi=(\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})\phi=(\frac{\delta\phi}{\delta x}\hat{i}+\frac{\delta\phi}{\delta y}\hat{j}+\frac{\delta\phi}{\delta z}\hat{k})=\hat{i}\frac{\delta}{\delta x}\phi+\hat{j}\frac{\delta}{\delta y}\phi+\hat{k}\frac{\delta}{\delta z}\phi$$

$$=\frac{1}{2}\hat{i}\frac{\delta}{\delta x}\ln(x^2+y^2+z^2)+\frac{1}{2}\hat{j}\frac{\delta}{\delta y}\ln(x^2+y^2+z^2)+\frac{1}{2}\hat{k}\frac{\delta}{\delta z}\ln(x^2+y^2+z^2)$$

$$=\frac{1}{2}\hat{i}\,(\frac{1}{x^2+y^2+z^2})\,\{\frac{\delta}{\delta x}(x^2+y^2+z^2)\}+\frac{1}{2}\hat{j}\,(\frac{1}{x^2+y^2+z^2})\,\{\frac{\delta}{\delta y}(x^2+y^2+z^2)\}+\frac{1}{2}\hat{k}\,(\frac{1}{x^2+y^2+z^2})\,\{\frac{\delta}{\delta z}(x^2+y^2+z^2)\}$$

$$=\frac{1}{2}\hat{i}\,(\frac{1}{x^2+y^2+z^2})\,(2x+0+0)+\frac{1}{2}\hat{j}\,(\frac{1}{x^2+y^2+z^2})\,(0+2y+0)+\frac{1}{2}\hat{k}\,(\frac{1}{x^2+y^2+z^2})\,(0+0+2z)$$

$$=\frac{1}{2}\hat{i}\,(\frac{2x}{x^2+y^2+z^2})+\frac{1}{2}\hat{j}\,(\frac{2y}{x^2+y^2+z^2})+\frac{1}{2}\hat{k}\,(\frac{2z}{x^2+y^2+z^2})$$

$$= \frac{1}{2}\times 2\{\frac{x\hat{i}+y\hat{j}+z\hat{k}}{x^2+y^2+z^2}\} = \{\frac{x\hat{i}+y\hat{j}+z\hat{k}}{x^2+y^2+z^2}\} = \frac{\vec{r}}{|\vec{r}|^2}$$ Answer

(b) Given, $\phi = \frac{1}{|\vec{r}|} = \frac{1}{\sqrt{x^2+y^2+z^2}} = (x^2+y^2+z^2)^{-1/2}$

$$\vec{\nabla}\phi = (\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})\phi = (\frac{\delta\phi}{\delta x}\hat{i}+\frac{\delta\phi}{\delta y}\hat{j}+\frac{\delta\phi}{\delta z}\hat{k}) = \hat{i}\frac{\delta}{\delta x}\phi+\hat{j}\frac{\delta}{\delta y}\phi+\hat{k}\frac{\delta}{\delta z}\phi$$

$$= \hat{i}\frac{\delta}{\delta x}(x^2+y^2+z^2)^{-1/2}+\hat{j}\frac{\delta}{\delta y}(x^2+y^2+z^2)^{-1/2}+\hat{k}\frac{\delta}{\delta z}(x^2+y^2+z^2)^{-1/2}$$

$$= \hat{i}(-\frac{1}{2})(x^2+y^2+z^2)^{-1/2-1}\frac{\delta}{\delta x}(x^2+y^2+z^2)\}+\hat{j}(-\frac{1}{2})$$
$$(x^2+y^2+z^2)^{-1/2-1}\frac{\delta}{\delta y}(x^2+y^2+z^2)\}+\hat{k}(-\frac{1}{2})(x^2+y^2+z^2)^{-1/2-1}$$
$$\frac{\delta}{\delta z}(x^2+y^2+z^2)\}$$

$$= \hat{i}(-\frac{1}{2})(x^2+y^2+z^2)^{-1/2-1}(2x+0+0)+\hat{j}(-\frac{1}{2})(x^2+y^2+z^2)^{-1/2-1}$$
$$(0+2y+0)+\hat{k}(-\frac{1}{2})(x^2+y^2+z^2)^{-1/2-1}\frac{\delta}{\delta z}(0+0+2z)$$

$$= \hat{i}(-x)(x^2+y^2+z^2)^{-3/2}+\hat{j}(-y)(x^2+y^2+z^2)^{-3/2}+\hat{k}(-z)(x^2+y^2+z^2)^{-3/2}$$

$$= \frac{-x\hat{i}}{(x^2+y^2+z^2)^{3/2}}+\frac{-y\hat{j}}{(x^2+y^2+z^2)^{3/2}}+\frac{-z\hat{k}}{(x^2+y^2+z^2)^{3/2}}$$

$$= -\frac{(x\hat{i}+y\hat{j}+z\hat{k})}{(x^2+y^2+z^2)^{3/2}} = -\frac{(x\hat{i}+y\hat{j}+z\hat{k})}{(x^2+y^2+z^2)^1.(x^2+y^2+z^2)^{1/2}}$$

$$= -\frac{\vec{r}}{|\vec{r}|^2.|\vec{r}|} = -\frac{\vec{r}}{|\vec{r}|^3}$$ Answer

Q#27: Find the level curve of $f(x,y) = -x^2+y^2$ passing through (2, 3). Draw Graph the gradient at the point (2, 3)

Answer: Given, $\mathbf{f(x,y) = -x^2+y^2}$

$$f(2,3) = -2^2+3^2 = -4+9 = 5$$

Hence the level curve is the hyperbola, i.e.

$$f(x,y) = -x^2+y^2 = 5$$

i.e. $-x^2+y^2 = 5$

i.e. $x^2 - y^2 = -5$

$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1$ [This is the equation of a hyperbola, i. e. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$] -----------------(i)

From (i),

$\Rightarrow \frac{x^2}{-5} - \frac{y^2}{-5} = 1 \qquad \Rightarrow x^2 - y^2 = -5 \qquad \Rightarrow y^2 = 5 + x^2 \qquad \Rightarrow y = \pm\sqrt{5 + x^2}$

$\Rightarrow y = \pm\sqrt{5 + x^2}$ ---------------------------(ii)

x	0	-1	-2	-3	1	2	3	-4	4	
$y = \pm\sqrt{5+x^2}$	$\pm\sqrt{5}$ $= \pm 2.23$	$\pm\sqrt{6}$ ± 2.44	± 3 ± 3	$\pm\sqrt{14}$ ± 3.74	$\pm\sqrt{6}$ ± 2.44	± 3 ± 3	$\pm\sqrt{14}$ ± 3.74	$\pm\sqrt{19}$ ± 4.35	$\pm\sqrt{19}$ ± 4.35	

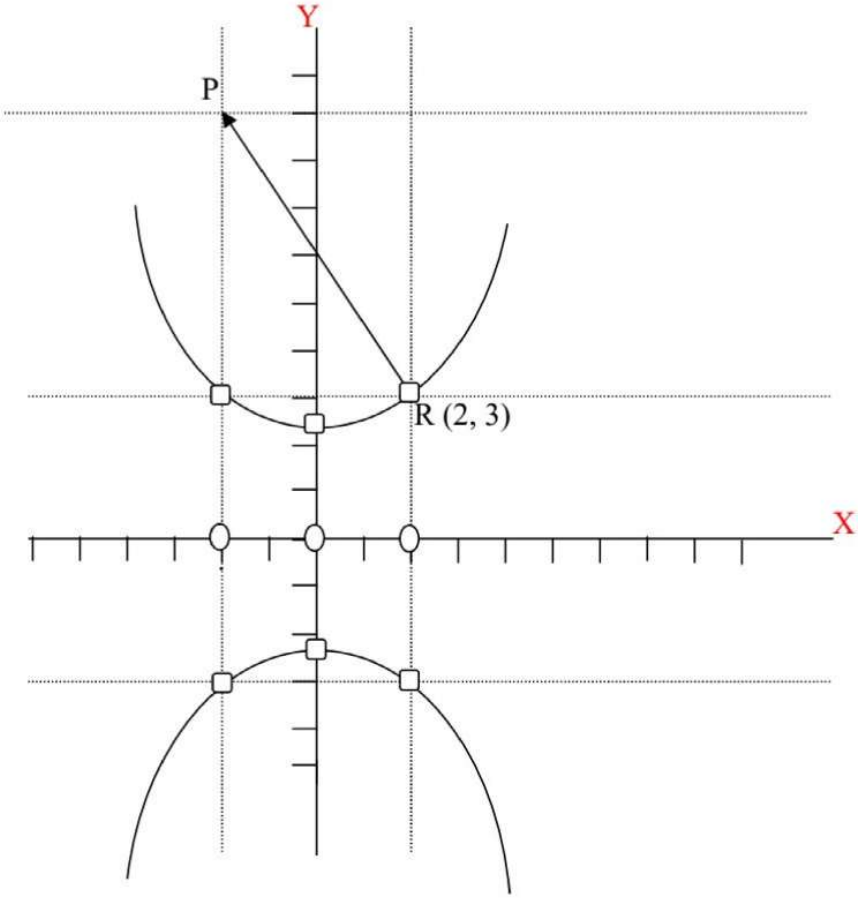

Figure # 69

Given, $f(x,y) = -x^2 + y^2$

Now, Gradient of the function, i.e.

$$\vec{\nabla} f(x,y) = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k})(-x^2 + y^2)$$

$$\vec{\nabla} f(x,y) = (\hat{i}\frac{\delta}{\delta x} + \hat{j}\frac{\delta}{\delta y} + \hat{k}\frac{\delta}{\delta z})(-x^2 + y^2)$$

$$\vec{\nabla} f(x,y) = \hat{i}\frac{\delta}{\delta x}(-x^2 + y^2) + \hat{j}\frac{\delta}{\delta y}(-x^2 + y^2) + \hat{k}\frac{\delta}{\delta z}(-x^2 + y^2)$$

$$\vec{\nabla} f(x,y) = \hat{i}\frac{\delta}{\delta x}(-x^2) + \hat{i}\frac{\delta}{\delta x}(y^2) + \hat{j}\frac{\delta}{\delta y}(-x^2) + \hat{j}\frac{\delta}{\delta y}(y^2) + \hat{k}\frac{\delta}{\delta z}(-x^2) + \hat{k}\frac{\delta}{\delta z}(y^2)$$

$$\vec{\nabla} f(x,y) = \hat{i}(-2x) + \hat{i}\times 0 + \hat{j}\times 0 + \hat{j}(2y) + \hat{k}\times 0 + \hat{k}\times 0$$

$$\vec{\nabla} f(x,y) = -2x\,\hat{i} + 2y\,\hat{j}$$

$$\vec{\nabla} \mathbf{f}(2,3) = -2\times 2\,\hat{\mathbf{i}} + 2\times 3\,\hat{\mathbf{j}}$$

$$\vec{\nabla} \mathbf{f}(2,3) = -4\,\hat{\mathbf{i}} + 6\,\hat{\mathbf{j}}$$

Hence the gradient Vector is $\vec{\mathbf{RP}} = \vec{\nabla}\mathbf{f}(2,3) = -4\,\hat{\mathbf{i}} + 6\,\hat{\mathbf{j}}$ the Answer

Q# 28: Sketch the level curve for the function $\mathbf{f(x,y) = x^2 + y^2}$ through the point (3, 4) and draw the gradient vector at this point.

Answer: Given, the function $\mathbf{f(x,y) = x^2 + y^2}$ through the point (3, 4),

$\mathbf{f(3,4) = 3^2 + 4^2}$

$\mathbf{f(3,4) = 9 + 16 = 25}$

Since $\mathbf{f(3,4) = 25}$, the level curve through the point $\mathbf{(3,4)}$ has the equation $\mathbf{f(x,y) = x^2 + y^2 = 25}$, which is the circle. That is $x^2 + y^2 = 25$ whose centre $\mathbf{(0,0)}$ and radius 5.

Now,

$\mathbf{f(x,y) = x^2 + y^2}$

Now, Gradient of the function,

$$\vec{\nabla}\mathbf{f(x,y)} = (\frac{\delta}{\delta x}\hat{\mathbf{i}} + \frac{\delta}{\delta y}\hat{\mathbf{j}} + \frac{\delta}{\delta z}\hat{\mathbf{k}})(x^2 + y^2)$$

$$\vec{\nabla}\mathbf{f(x,y)} = (\hat{\mathbf{i}}\frac{\delta}{\delta x} + \hat{\mathbf{j}}\frac{\delta}{\delta y} + \hat{\mathbf{k}}\frac{\delta}{\delta z})(x^2 + y^2)$$

$$\vec{\nabla}\mathbf{f(x,y)} = \hat{\mathbf{i}}\frac{\delta}{\delta x}(x^2 + y^2) + \hat{\mathbf{j}}\frac{\delta}{\delta y}(x^2 + y^2) + \hat{\mathbf{k}}\frac{\delta}{\delta z}(x^2 + y^2)$$

$$\vec{\nabla}\mathbf{f(x,y)} = \hat{\mathbf{i}}\frac{\delta}{\delta x}(x^2) + \hat{\mathbf{i}}\frac{\delta}{\delta x}(y^2) + \hat{\mathbf{j}}\frac{\delta}{\delta y}(x^2) + \hat{\mathbf{j}}\frac{\delta}{\delta y}(y^2) + \hat{\mathbf{k}}\frac{\delta}{\delta z}(x^2) + \hat{\mathbf{k}}\frac{\delta}{\delta z}(y^2)$$

$$\vec{\nabla} f(x,y) = \hat{i}(2x) + \hat{i}\times 0 + \hat{j}\times 0 + \hat{j}(2y) + \hat{k}\times 0 + \hat{k}\times 0$$

$$\vec{\nabla} f(x,y) = 2x\,\hat{i} + 2y\,\hat{j} \qquad \text{-------------------------(i)}$$

The gradient vector at $(3,4)$ is

$$\therefore \vec{\nabla} f(3,4) = 2\times 3\,\hat{i} + 2\times 4\,\hat{j}$$

$$\vec{\nabla} f(3,4) = 6\hat{i} + 8\hat{j} \qquad \text{-------------------------(ii)}$$

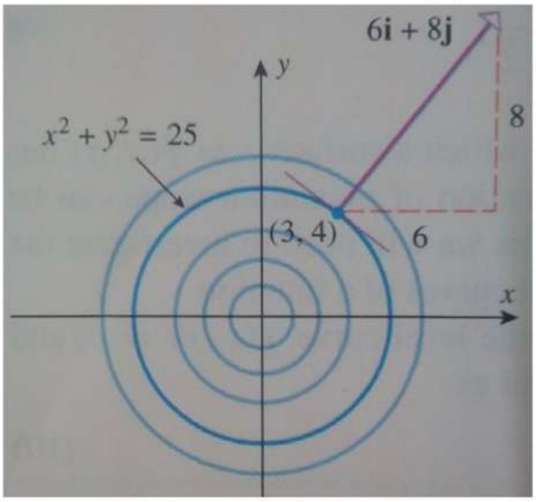

Figure # 70

Hence the gradient vector is perpendicular to the circle at $(3,4)$.

Q# 29: Sketch the gradient field of $\phi(x,y) = x + y$

Answer: Now, the gradient of the function $\phi(x,y) = x + y$

$$\vec{\nabla}\phi(x,y) = \left(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}\right)(x+y)$$

$$\vec{\nabla}\phi(x,y) = \left(\hat{i}\frac{\delta}{\delta x} + \hat{j}\frac{\delta}{\delta y} + \hat{k}\frac{\delta}{\delta z}\right)(x+y)$$

$$\vec{\nabla}\phi(x,y) = \hat{i}\frac{\delta}{\delta x}(x+y) + \hat{j}\frac{\delta}{\delta y}(x+y) + \hat{k}\frac{\delta}{\delta z}(x+y)$$

$$\vec{\nabla}\phi(x,y) = \hat{i}(1+0) + \hat{j}(0+1) + \hat{k}(0+0)$$

$$\vec{\nabla}\phi(x,y) = \hat{i} + \hat{j}$$

This is the same at each point. A portion of the vector field is sketched in figure below:

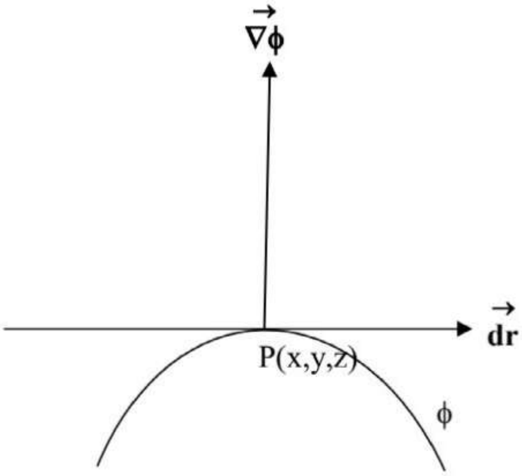

Figure # 73

It is clear that the vector $\vec{\nabla}\phi$ is perpendicular (normal) to the tangent vector $\vec{dr}$ at a point $\mathbf{P(x,y,z)}$ that is $\vec{\nabla}\phi . d\vec{r} = 0$

So we conclude that $\vec{\nabla}\phi$ is normal (perpendicular) vector to the surface $\phi(x,y,z) = c$ at (x,y,z).

[N.B. We always remember that $\vec{\nabla}\phi$ is perpendicular to the tangent to the surface but not with surface directly and $\vec{\nabla}\phi$ is normal to the surface $\phi(x,y,z) = c$, Where $\vec{\nabla}\phi$ is a vector component that is $\vec{\nabla}\phi = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\phi$]

Q# 32: Find a unit normal to the surface $x^2y + 2xz = 4$ at the point $(2,-2,3)$

Answer: Given, $\phi(x,y,z) = x^2y + 2xz = 4$

$$\therefore \vec{\nabla}\phi = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k})\phi = (\frac{\delta\phi}{\delta x}\hat{i} + \frac{\delta\phi}{\delta y}\hat{j} + \frac{\delta\phi}{\delta z}\hat{k}) = \hat{i}\frac{\delta}{\delta x}\phi + \hat{j}\frac{\delta}{\delta y}\phi + \hat{k}\frac{\delta}{\delta z}\phi$$

$$= \hat{i}\frac{\delta}{\delta x}(x^2y + 2xz) + \hat{j}\frac{\delta}{\delta y}(x^2y + 2xz) + \hat{k}\frac{\delta}{\delta z}(x^2y + 2xz)$$

$$= (2xy + 2z)\hat{i} + x^2\hat{j} + 2x\hat{k}$$

$$= (2\times2\times(-2) + 2\times3)\hat{i} + 2^2\hat{j} + 2\times2\hat{k} \text{ at the point } (2,-2,3)$$

$$= -2\hat{i} + 4\hat{j} + 4\hat{k}$$

Then a unit normal to the surface $= \dfrac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{\sqrt{(-2)^2 + (4)^2 + (4)^2}} = -\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$ Answer

Q# 33:

Find the level surface of $F(x,y,z)=x^2+y^2+z^2$ passing through $(1,1,1)$. Graph the gradient at the point.

Answer: Given, $F(x,y,z)=x^2+y^2+z^2$

$$\therefore F(1,1,1)=1^2+1^2+1^2=3$$

Hence $F(x,y,z)=x^2+y^2+z^2=3$

Because $F(1,1,1)=3$, **the level surface passing through (1,1,1) is the sphere** $x^2+y^2+z^2=3$.

The gradient of the function is

$F(x,y,z)=x^2+y^2+z^2$

$$F(x,y,z)=x^2+y^2+z^2$$

$$\therefore \nabla F(x,y,z)=(\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})F$$

$$\therefore \nabla F(x,y,z)=(\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})(x^2+y^2+z^2)$$

$$\therefore \nabla F(x,y,z)=$$

$$\hat{i}\frac{\delta}{\delta x}(x^2+y^2+z^2)+\hat{j}\frac{\delta}{\delta y}(x^2+y^2+z^2)+\hat{k}\frac{\delta}{\delta z}(x^2+y^2+z^2)$$

$$\therefore \nabla F(x,y,z)=\hat{i}\frac{\delta}{\delta x}(2x+0+0)+\hat{j}\frac{\delta}{\delta y}(0+2y+0)+\hat{k}\frac{\delta}{\delta z}(0+0+2z)$$

$$\therefore \nabla F(x,y,z)=\hat{i}(2x)+\hat{j}(2y)+\hat{k}(2z)$$

$$\therefore \nabla F(x,y,z)=2x\hat{i}+2y\hat{j}+2z\hat{k}$$

And so, at the given point

$$\therefore \nabla F(1,1,1)=2.1\hat{i}+2.1\hat{j}+2.1\hat{k}$$

$$\therefore \nabla F(1,1,1)=2\hat{i}+2\hat{j}+2\hat{k}$$

The level surface and $\nabla F(1,1,1)$ are illustrated in figure no 74

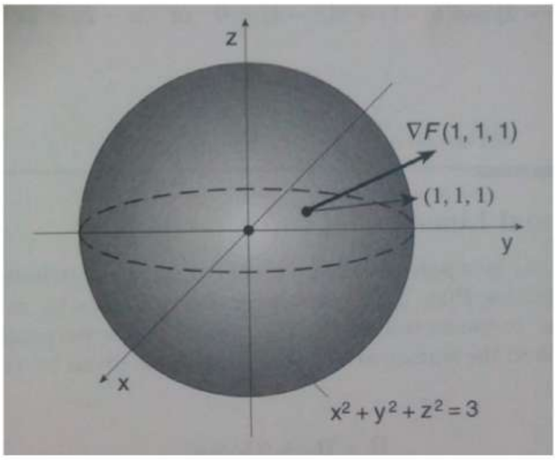

Figure # 74

Q#34: Prove that the angle between the surfaces at the point is equal to the angle between the normals to the surfaces at the point.

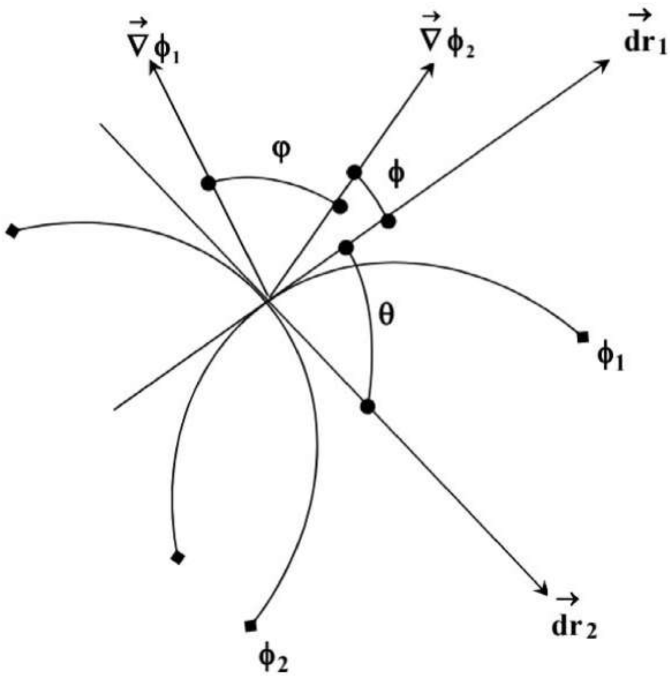

Figure # 75

Here, $\hat{\eta}_1$ is the unit vector of $\vec{\nabla}\phi_1$ and $\hat{\eta}_2$ is the unit vector of $\vec{\nabla}\phi_2$

We can write,

$$\hat{\eta}_1 = \frac{\vec{\nabla}\phi_1}{\left|\vec{\nabla}\phi_1\right|} \text{ and } \hat{\eta}_2 = \frac{\vec{\nabla}\phi_2}{\left|\vec{\nabla}\phi_2\right|}$$

We have,

$$\vec{a}.\vec{b} = \left|\vec{a}\right|\left|\vec{b}\right|\cos\theta$$

$$\therefore \hat{\eta}_1 . \hat{\eta}_2 = \left|\hat{\eta}_1\right|\left|\hat{\eta}_2\right|\cos\varphi \qquad [\because \varphi \text{ be the angle between the normals to the surfaces } \phi_1 \text{ and } \phi_2]$$

$$\therefore \hat{\eta}_1 . \hat{\eta}_2 = 1.1\cos\varphi \qquad [\because \text{The length or magnitude of unit vector is 1}]$$

$$\Rightarrow \frac{\vec{\nabla}\phi_1}{\left|\vec{\nabla}\phi_1\right|} . \frac{\vec{\nabla}\phi_2}{\left|\vec{\nabla}\phi_2\right|} = 1.1\cos\varphi$$

$$\Rightarrow \vec{\nabla}\phi_1 . \vec{\nabla}\phi_2 = \left|\vec{\nabla}\phi_1\right|\left|\vec{\nabla}\phi_2\right|\cos\varphi \text{-------------------------(i)}$$

Again, from figure # 75

$$\varphi + \phi = 90$$
$$\underline{\pm\theta \pm \phi = \pm 90}$$
$$\varphi - \theta = 0$$
$$\therefore \varphi = \theta \text{---(ii)}$$ ***(Proved)***

Q# 35: Find the angle between the surfaces $\mathbf{x^2 + y^2 + z^2 = 9}$ and $\mathbf{z = x^2 + y^2 - 3}$ at the point (2,-1, 2)

Answer:

Given, $z = x^2 + y^2 - 3$

$$\Rightarrow x^2 + y^2 - z = 3$$

Let, $\phi_1(x,y,z) = x^2 + y^2 + z^2 = 9$ and $\phi_2(x,y,z) = x^2 + y^2 - z = 3$

$$\therefore \nabla\phi_1 = \left(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}\right)\phi_1 = \left(\frac{\delta\phi_1}{\delta x}\hat{i} + \frac{\delta\phi_1}{\delta y}\hat{j} + \frac{\delta\phi_1}{\delta z}\hat{k}\right) = \hat{i}\frac{\delta}{\delta x}\phi_1 + \hat{j}\frac{\delta}{\delta y}\phi_1 + \hat{k}\frac{\delta}{\delta z}\phi_1$$

$$= \hat{i}\frac{\delta}{\delta x}(x^2 + y^2 + z^2) + \hat{j}\frac{\delta}{\delta y}(x^2 + y^2 + z^2) + \hat{k}\frac{\delta}{\delta z}(x^2 + y^2 + z^2)$$

$$= 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$= 4\hat{i} - 2\hat{j} + 4\hat{k} \text{ at the point } (2,-1,2)$$

A normal to $x^2+y^2+z^2=9$ at $(2,-1,2)$ is $\nabla\phi_1 = 4\hat{i}-2\hat{j}+4\hat{k}$

Again,

$\phi_2(x,y,z) = x^2+y^2-z=3$

$$\therefore \nabla\phi_2 = (\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})\phi_2 = (\frac{\delta\phi_2}{\delta x}\hat{i}+\frac{\delta\phi_2}{\delta y}\hat{j}+\frac{\delta\phi_2}{\delta z}\hat{k}) = \hat{i}\frac{\delta}{\delta x}\phi_2+\hat{j}\frac{\delta}{\delta y}\phi_2+\hat{k}\frac{\delta}{\delta z}\phi_2$$

$$= \hat{i}\frac{\delta}{\delta x}(x^2+y^2-z)+\hat{j}\frac{\delta}{\delta y}(x^2+y^2-z)+\hat{k}\frac{\delta}{\delta z}(x^2+y^2-z)$$

$$= 2x\hat{i}+2y\hat{j}-\hat{k}$$

$$= 4\hat{i}-2\hat{j}-\hat{k} \text{ at the point } (2,-1,2)$$

A normal to $x^2+y^2-z=3$ at $(2,-1,2)$ is $\nabla\phi_2 = 4\hat{i}-2\hat{j}-\hat{k}$

Let φ be the angle between the normals to the surfaces at the point $(2,-1,2)$

Then we have,

$\nabla\phi_1 . \nabla\phi_2 = |\nabla\phi_1|\,|\nabla\phi_2|\cos\varphi$, **[from equation no (i), page no 78]**

$$\Rightarrow (4\hat{i}-2\hat{j}+4\hat{k}).(4\hat{i}-2\hat{j}-\hat{k}) = \left|4\hat{i}-2\hat{j}+4\hat{k}\right|\left|4\hat{i}-2\hat{j}-\hat{k}\right|\cos\varphi$$

$$\Rightarrow 16+4-4 = \sqrt{(4)^2+(-2)^2+(4)^2}\sqrt{(4)^2+(-2)^2+(-1)^2}\cos\varphi$$

$$\Rightarrow 16 = 6\sqrt{21}\cos\varphi$$

$$\Rightarrow \cos\varphi = \frac{16}{6\sqrt{21}}$$

$$\therefore \varphi = \cos^{-1}(\frac{16}{6\sqrt{21}}) \text{ Answer } [\because \varphi = \theta]$$

Q# 36: Prove that $\underline{\nabla}.[\frac{f(r)}{r}\vec{r}] = \frac{1}{r^2}\frac{d}{dr}[r^2 f(r)]$

Here $\left|\vec{r}\right| = r$

L.H.S.

$$\underline{\nabla}.[\frac{f(r)}{r}\vec{r}]$$

$$= (\hat{i}\frac{\delta}{\delta x}+\hat{j}\frac{\delta}{\delta y}+\hat{k}\frac{\delta}{\delta z}).[\frac{f(r)}{r}(x\hat{i}+y\hat{j}+z\hat{k})] \quad [\because \vec{r} = x\hat{i}+y\hat{j}+z\hat{k}]$$

$$= (\hat{i}\frac{\delta}{\delta x}+\hat{j}\frac{\delta}{\delta y}+\hat{k}\frac{\delta}{\delta z}).(\hat{i}\frac{f(r)}{r}x+\hat{j}\frac{f(r)}{r}y+\hat{k}\frac{f(r)}{r}z)$$

$$= \frac{\delta}{\delta x}\frac{f(r)}{r}x+\frac{\delta}{\delta y}\frac{f(r)}{r}y+\frac{\delta}{\delta z}\frac{f(r)}{r}z \text{ ------------------------------------(i)}$$

$$[\because \hat{i}.\hat{i} = 1; \hat{j}.\hat{j} = 1; \hat{k}.\hat{k} = 1]$$

direction of $\hat{a}$ and $\frac{d\phi}{ds} = \hat{a}$. Grad ϕ will be a maximum when $\hat{a}$ and grad ϕ have the same direction, since then, $\hat{a} \cdot grad\ \phi = |\hat{a}||grad\ \phi| \cos\theta$ and θ will be zero

Thus direction of grad ϕ gives the direction in which the maximum rate of change of ϕ occurs.

Q# 37: Find the directional derivative of the function $\phi = x^2z + 2xy^2 + yz^2$ at the point of (1, 2,-1) in the direction of the vector $\vec{A} = 2\hat{i} + 3\hat{j} + \hat{k}$.

We start off with $\phi = x^2z + 2xy^2 + yz^2$

$$\therefore \nabla\phi = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k})\phi$$

$$\therefore \nabla\phi = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k})(x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla\phi =$$

$$\hat{i}\frac{\delta}{\delta x}(x^2z + 2xy^2 + yz^2) + \hat{j}\frac{\delta}{\delta y}(x^2z + 2xy^2 + yz^2) + \hat{k}\frac{\delta}{\delta z}(x^2z + 2xy^2 + yz^2)$$

$$\therefore \nabla\phi = \hat{i}(2xz + 2.1.y^2 + 0) + \hat{j}(0 + 2x.2y + 1.z^2) + \hat{k}(x^2.1 + 0 + y.2z)$$

$$\therefore \nabla\phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

At the point $(1,2,-1)$

$$\nabla\phi = \hat{i}(2xz + 2y^2) + \hat{j}(4xy + z^2) + \hat{k}(x^2 + 2yz)$$

$$\therefore \nabla\phi = \hat{i}[2.1.(-1) + 2(2^2)] + \hat{j}[4.1.2 + (-1)^2] + \hat{k}[(1^2 + 2.2.(-1)]$$

$$\therefore \nabla\phi = \hat{i}[-2 + 8] + \hat{j}[8 + 1)] + \hat{k}[(1 - 4]$$

$$\therefore \nabla\phi = \hat{i}[6] + \hat{j}[9)] + \hat{k}(-3)$$

$$\therefore \nabla\phi = 6\hat{i} + 9\hat{j} - 3\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|A|}$$

$$\therefore \hat{a} = \frac{2\hat{i} + 3\hat{j} - 4\hat{k}}{|2^2 + 3^2 + (-4)^2|}$$

$$\therefore \hat{a} = \frac{2\hat{i}+3\hat{j}-4\hat{k}}{|4+9+16|}$$

$$\therefore \hat{a} = \frac{2\hat{i}+3\hat{j}-4\hat{k}}{|29|}$$

Hence $\dfrac{d\phi}{ds} = \hat{a}.\vec{\nabla}\phi$

$$\frac{d\phi}{ds} = \frac{2\hat{i}+3\hat{j}-4\hat{k}}{|29|}.(6\hat{i}+9\hat{j}-3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|}(2\hat{i}+3\hat{j}-4\hat{k}).(6\hat{i}+9\hat{j}-3\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{|29|}(12+27+12)$$

$$\frac{d\phi}{ds} = \frac{51}{|29|} \; \textit{Answer}$$

Q#38: Find the directional derivative of the function $\phi = x^2y + y^2z + z^2x$ at the point of (1, -1, 2) in the direction of the vector $\vec{A} = 4\hat{i}+2\hat{j}-5\hat{k}$.

We have $\quad \phi = x^2y + y^2z + z^2x$

$$\therefore \nabla\phi = (\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})\phi$$

$$\therefore \nabla\phi = (\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k})(x^2y + y^2z + z^2x)$$

$$\therefore \nabla\phi = \hat{i}\frac{\delta}{\delta x}(x^2y + y^2z + z^2x)+\hat{j}\frac{\delta}{\delta y}(x^2y + y^2z + z^2x)+\hat{k}\frac{\delta}{\delta z}(x^2y + y^2z + z^2x)$$

$$\therefore \nabla\phi = \hat{i}(2xy+0+z^2.1)+\hat{j}(x^2.1+2yz+0)+\hat{k}(0+y^2.1+2zx)$$

$$\therefore \nabla\phi = \hat{i}(2xy+z^2)+\hat{j}(x^2+2yz)+\hat{k}(y^2+2zx)$$

At the point $(1,-1,2)$

$$\therefore \nabla\phi = \hat{i}(2xy+z^2)+\hat{j}(x^2+2yz)+\hat{k}(y^2+2zx)$$

$$\therefore \nabla\phi = \hat{i}[2.1.(-1)+2^2)]+\hat{j}[1^2+2(-1).2]+\hat{k}[(-1)^2+2.2.1]$$

$$\therefore \nabla\phi = \hat{i}[-2+4]+\hat{j}[1-4]+\hat{k}[1+4]$$

$$\therefore \nabla\phi = 2\hat{i}-3\hat{j}+5\hat{k}$$

Next we have to find out the unit vector $\hat{a}$ where $\vec{A} = 4\hat{i}+2\hat{j}-5\hat{k}$

$$\therefore \hat{a} = \frac{\vec{A}}{|A|}$$

$$\therefore \hat{a} = \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{\sqrt{4^2 + 2^2 + (-5)^2}}$$

$$\therefore \hat{a} = \frac{4\hat{i} + 2\hat{j} - 5\hat{k}}{\sqrt{45}}$$

Hence $\dfrac{d\phi}{ds} = \hat{a} . \vec{\nabla}\phi$

$$\frac{d\phi}{ds} = \frac{(4\hat{i} + 2\hat{j} - 5\hat{k})}{\sqrt{45}} . (2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{\sqrt{45}} (4\hat{i} + 2\hat{j} - 5\hat{k}) . (2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\frac{d\phi}{ds} = \frac{1}{\sqrt{45}} (8 - 6 - 25)$$

$$\frac{d\phi}{ds} = \frac{1}{\sqrt{45}} (-23) \text{ } \textit{Answer}$$

Q#39: Find the directional derivative of the function $\phi = (x, y, z) = x^2 - y^2 + 2z^2$ at the point of (1,2,3) in the direction of the vector $\vec{A} = 4\hat{i} - 2\hat{j} + \hat{k}$.

Answer: Let, $\phi(x, y, z) = x^2 - y^2 + 2z^2$

$$\frac{\delta\phi}{\delta x} = \frac{\delta}{\delta x}(x^2 - y^2 + 2z^2)$$

$$\frac{\delta\phi}{\delta x} = (2x - 0 + 0)$$

$$\frac{\delta\phi}{\delta x} = 2x$$

$$\phi = (x, y, z) = x^2 - y^2 + 2z^2$$

$$\frac{\delta\phi}{\delta y} = \frac{\delta}{\delta y}(x^2 - y^2 + 2z^2)$$

$$\frac{\delta\phi}{\delta y} = (0 - 2y + 0)$$

$$\frac{\delta\phi}{\delta y} = -2y$$

$$\phi(x, y, z) = x^2 - y^2 + 2z^2$$

Q# 41: If $\vec{V}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ Find divergence of $\vec{V}$ that is $\text{div}\,\vec{V}$

Answer: $\text{div}\,\vec{V} = \nabla.\vec{V} = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}).\,(xz\hat{i} + xyz\hat{j} - y^2\hat{k})$

$$= \frac{\delta}{\delta x}(xz) + \frac{\delta}{\delta y}(xyz) - \frac{\delta}{\delta z}(y^2) \quad [\therefore \hat{i}.\hat{i} = 1,\ \hat{j}.\hat{j} = 1, \hat{k}.\hat{k} = 1, \hat{k}.\hat{i} = 0 \text{ etc.}]$$

$= z + xz$ Answer

Q# 42: Let $\vec{V}$ be a constant vector field. Show that $\text{div}\,\vec{V} = 0$

Answer: Let, $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$, where a,b,c are constants, Then

$$\text{div}\,\vec{V} = \nabla.\vec{V} = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}).\,(a\hat{i} + b\hat{j} + c\hat{k})$$

$$= \frac{\delta}{\delta x}(a) + \frac{\delta}{\delta y}(b) + \frac{\delta}{\delta z}(c) \qquad [\therefore \hat{i}.\hat{i} = 1,\ \hat{j}.\hat{j} = 1, \hat{k}.\hat{k} = 1, \hat{k}.\hat{i} = 0 \text{ etc.}]$$

$= 0$

Q# 43: what is solenoidal?

Answer: If $\vec{A}$ is solenoidal then $\vec{\nabla}.\vec{A} = 0$

Home Task: Determine the constant a so that the vector

$\vec{F} = (x+2y)\hat{i} + (y-2z)\hat{j} + (x+az)\hat{k}$ is solenoidal

Q# 44:Show that the vector field $\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2+y^2}}$ is a "sink field". Plot and give a physical interpretation. [Here V is a unit vector; Since it is divided by its length]

Answer: given,

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2+y^2}}$$

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{(-x)^2+(-y)^2}}$$

$$\vec{v} = \frac{-x\hat{i} - y\hat{j}}{\sqrt{x^2+y^2}}$$

$$\vec{v} = \frac{-x}{\sqrt{x^2+y^2}}\hat{i} + \frac{-y}{\sqrt{x^2+y^2}}\hat{j}$$

$$\therefore \operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv (\frac{\delta}{\delta x}\hat{i}+\frac{\delta}{\delta y}\hat{j}+\frac{\delta}{\delta z}\hat{k}).\ (\frac{-x}{\sqrt{x^2+y^2}}\hat{i}+\frac{-y}{\sqrt{x^2+y^2}}\hat{j}+0.\hat{k})$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{\delta}{\delta x}(\frac{-x}{\sqrt{x^2+y^2}})+\frac{\delta}{\delta y}(\frac{-y}{\sqrt{x^2+y^2}})$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{\delta}{\delta x}(\frac{-x}{\sqrt{x^2+y^2}})+\frac{\delta}{\delta y}(\frac{-y}{\sqrt{x^2+y^2}})$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{(\sqrt{x^2+y^2})\frac{\delta}{\delta x}(-x)-(-x)\frac{\delta}{\delta x}(\sqrt{x^2+y^2})}{(\sqrt{x^2+y^2})^2}+\frac{(\sqrt{x^2+y^2})\frac{\delta}{\delta y}(-y)-(-y)\frac{\delta}{\delta y}(\sqrt{x^2+y^2})}{(\sqrt{x^2+y^2})^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{(\sqrt{x^2+y^2})(-1)+x\frac{\delta}{\delta x}(\sqrt{x^2+y^2})}{(\sqrt{x^2+y^2})^2}+\frac{(\sqrt{x^2+y^2})(-1)+y\frac{\delta}{\delta y}(\sqrt{x^2+y^2})}{(\sqrt{x^2+y^2})^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(\sqrt{x^2+y^2})+x\frac{\delta}{\delta x}(\sqrt{x^2+y^2})}{(\sqrt{x^2+y^2})^2}+\frac{-(\sqrt{x^2+y^2})+y\frac{\delta}{\delta y}(\sqrt{x^2+y^2})}{(\sqrt{x^2+y^2})^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(\sqrt{x^2+y^2})+x\frac{\delta}{\delta x}(x^2+y^2)^{1/2}}{(\sqrt{x^2+y^2})^2}+\frac{-(\sqrt{x^2+y^2})+y\frac{\delta}{\delta y}(x^2+y^2)^{1/2}}{(\sqrt{x^2+y^2})^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(\sqrt{x^2+y^2})+x\frac{1}{2}(x^2+y^2)^{\frac{1}{2}-1}.\frac{\delta}{\delta x}(x^2+y^2)}{(\sqrt{x^2+y^2})^2}+\frac{-(\sqrt{x^2+y^2})+y\frac{1}{2}(x^2+y^2)^{\frac{1}{2}-1}.\frac{\delta}{\delta y}(x^2+y^2)}{(\sqrt{x^2+y^2})^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(\sqrt{x^2+y^2})+x\frac{1}{2}(x^2+y^2)^{-\frac{1}{2}}.(2x)}{(\sqrt{x^2+y^2})^2}+\frac{-(\sqrt{x^2+y^2})+y\frac{1}{2}(x^2+y^2)^{-\frac{1}{2}}.(2y)}{(\sqrt{x^2+y^2})^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(\sqrt{x^2+y^2})+x^2(x^2+y^2)^{-\frac{1}{2}}}{(\sqrt{x^2+y^2})^2}+\frac{-(\sqrt{x^2+y^2})+y^2(x^2+y^2)^{-\frac{1}{2}}}{(\sqrt{x^2+y^2})^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(\sqrt{x^2+y^2})+\dfrac{x^2}{(x^2+y^2)^{\frac{1}{2}}}}{x^2+y^2}+\frac{-(\sqrt{x^2+y^2})+\dfrac{y^2}{(x^2+y^2)^{\frac{1}{2}}}}{x^2+y^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(\sqrt{x^2+y^2})+\dfrac{x^2}{\sqrt{x^2+y^2}}}{x^2+y^2}+\frac{-(\sqrt{x^2+y^2})+\dfrac{y^2}{\sqrt{x^2+y^2}}}{x^2+y^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{\frac{-x^2-y^2+x^2}{\sqrt{x^2+y^2}}}{x^2+y^2} + \frac{\frac{-x^2-y^2+y^2}{\sqrt{x^2+y^2}}}{x^2+y^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{\frac{-y^2}{\sqrt{x^2+y^2}}}{x^2+y^2} + \frac{\frac{-x^2}{\sqrt{x^2+y^2}}}{x^2+y^2}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-y^2}{(x^2+y^2)(\sqrt{x^2+y^2})} + \frac{-x^2}{(x^2+y^2)(\sqrt{x^2+y^2})}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-y^2-x^2}{(x^2+y^2)(\sqrt{x^2+y^2})}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-(x^2+y^2)}{(x^2+y^2)(\sqrt{x^2+y^2})}$$

$$\operatorname{div}\vec{v} \equiv \vec{\nabla}.\vec{v} \equiv \frac{-1}{\sqrt{x^2+y^2}} < 0$$

So, $\vec{v}$ a "sink field"

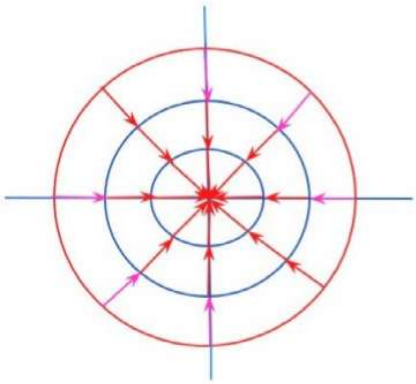

Figure # 78

Q# 45: Let $\vec{V}$ be a constant vector field. Show that $\operatorname{curl}\vec{V} = 0$

Answer Let, $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$, **where a, b, c are constants, Then**

$$\operatorname{curl}\vec{V} = \nabla \times \vec{V} = \left(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}\right) \times (a\hat{i} + b\hat{j} + c\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ a & b & c \end{vmatrix}$$

$$= \hat{i}(\frac{\delta c}{\delta y} - \frac{\delta b}{\delta z}) - \hat{j}(\frac{\delta c}{\delta x} - \frac{\delta a}{\delta z}) + \hat{k}(\frac{\delta b}{\delta x} - \frac{\delta a}{\delta y})$$

$$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0-0)$$

$$= 0$$

Q# 46: If $\vec{v}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ **Find** $\text{curl}\,\vec{V}$

Answer: The curl of a vector field $\vec{v}(x,y,z) = xz\hat{i} + xyz\hat{j} - y^2\hat{k}$ is defined by

$$\nabla \times \vec{v} = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}) \times (xz\hat{i} + xyz\hat{j} - y^2\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ xz & xyz & -y^2 \end{vmatrix}$$

$$= \hat{i}[\frac{\delta}{\delta y}(-y^2) - \frac{\delta}{\delta z}(xyz)] - \hat{j}[\frac{\delta}{\delta x}(-y^2) - \frac{\delta}{\delta z}(xz)] + \hat{k}[\frac{\delta}{\delta x}(xyz) - \frac{\delta}{\delta y}(xz)]$$

$$= \hat{i}[-2y - xy] - \hat{j}[0 - x] + \hat{k}[yz - 0]$$

$$= -[2y + xy]\hat{i} + \hat{j}x + \hat{k}yz$$

$$= -[2y + xy]\hat{i} + x\hat{j} + yz\hat{k}$$

Q# 47: What is i**rrotational Field or conservative vector field?**

A vector field $\vec{V}$ for which the curl vanishes, that is: $\vec{V} \times \vec{V} = 0$

Q# 48: Determine $\vec{F}$ is a conservative vector field or not where

$\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$

Answer:

So all that we need to do is compute the curl and see if we get the zero vector or not.

The curl of a vector field $\vec{F} = x^2y\hat{i} + xyz\hat{j} - x^2y^2\hat{k}$ is defined by

$$\nabla \times \vec{F} = \left(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}\right) \times (x^2y\,\hat{i} + xyz\,\hat{j} - x^2y^2\,\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ x^2y & xyz & -x^2y^2 \end{vmatrix}$$

$$= \hat{i}\left[\frac{\delta}{\delta y}(-x^2y^2) - \frac{\delta}{\delta z}(xyz)\right] - \hat{j}\left[\frac{\delta}{\delta x}(-x^2y^2) - \frac{\delta}{\delta z}(x^2y)\right]$$

$$+ \hat{k}\left[\frac{\delta}{\delta x}(xyz) - \frac{\delta}{\delta y}(x^2y)\right]$$

$$= \hat{i}[-2x^2y - xy] - \hat{j}[-2xy^2 - 0] + \hat{k}[yz - x^2]$$

$$\neq 0$$

So, the curl isn't the zero vectors and so this vector field is not conservative.

Home Task

Find the values of constants a, b and c so that,

$\vec{F} = (x + 2y + az)\,\hat{i} + (bx - 3y - z)\,\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational.

Q# 49: If $\vec{A} = (xz^3\,\hat{i} - 2x^2yz\,\hat{j} + 2yz^4\,\hat{k})$, find $\nabla \times \vec{A}$ (or curl A) at the point (1,-1, 1).

Answer:

$$\nabla \times \vec{A} = \left(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}\right) \times (xz^3\,\hat{i} - 2x^2yz\,\hat{j} + 2yz^4\,\hat{k})$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\delta}{\delta x} & \frac{\delta}{\delta y} & \frac{\delta}{\delta z} \\ xz^3 & -2x^2yz & 2yz^4 \end{vmatrix}$$

$$= \hat{i}\left[\frac{\delta}{\delta y}(2yz^4) - \frac{\delta}{\delta z}(-2x^2yz)\right] - \hat{j}\left[\frac{\delta}{\delta x}(2yz^4) - \frac{\delta}{\delta z}(xz^3)\right] + \hat{k}\left[\frac{\delta}{\delta x}(-2x^2yz) - \frac{\delta}{\delta y}(xz^3)\right]$$

$$= \hat{i}[2z^4 + 2x^2y] - \hat{j}[0 - 3xz^2] + \hat{k}[-4xyz - 0]$$

$$= \hat{i}[2.1^4 + 2.1^2.(-1)] - \hat{j}[0 - 3.1.1^2] + \hat{k}[-4.1.(-1).1 - 0]$$

$$= \hat{i}[2 - 2] - \hat{j}[0 - 3] + \hat{k}[4]$$

$$= 3\,\hat{j} + 4\hat{k}$$

Q# 55: Show that $\vec{r}.\vec{r} = r^2$

Also,
$$\vec{r}.\vec{r} = (x\hat{i} + y\hat{j} + z\hat{k}).(x\hat{i} + y\hat{j} + z\hat{k})$$
$$\vec{r}.\vec{r} = x^2 + y^2 + z^2 \; [\because \hat{i}.\hat{i} = 1; \hat{j}.\hat{j} = 1; \hat{k}.\hat{k} = 1]$$
$$= r^2 \; [\because r^2 = x^2 + y^2 + z^2]$$

Similarly,

$\vec{A}.\vec{A}. = A^2$

$\vec{\nabla}.\vec{\nabla} = \nabla^2$

Q# 56: Show that, $\vec{\nabla} = \frac{\vec{r}}{r}\frac{\partial}{\partial r}$

Proof: **L.H.S** $= \vec{\nabla}$
$$= \hat{i}\frac{\partial}{\partial x} + \hat{j}\frac{\partial}{\partial y} + \hat{k}\frac{\partial}{\partial z}$$
$$= \hat{i}\frac{\partial}{\partial x}\frac{\partial r}{\partial r} + \hat{j}\frac{\partial}{\partial y}\frac{\partial r}{\partial r} + \hat{k}\frac{\partial}{\partial z}\frac{\partial r}{\partial r}$$
$$= \hat{i}\frac{\partial r}{\partial x}\frac{\partial}{\partial r} + \hat{j}\frac{\partial r}{\partial y}\frac{\partial}{\partial r} + \hat{k}\frac{\partial r}{\partial z}\frac{\partial}{\partial r}$$
$$= \left(\hat{i}\frac{\partial r}{\partial x} + \hat{j}\frac{\partial r}{\partial y} + \hat{k}\frac{\partial r}{\partial z}\right)\frac{\partial}{\partial r}$$
$$= \left(\hat{i}\frac{x}{r} + \hat{j}\frac{y}{r} + \hat{k}\frac{z}{r}\right)\frac{\partial}{\partial r} \; [\because \frac{\partial r}{\partial x} = \frac{x}{r}; \frac{\partial r}{\partial y} = \frac{y}{r}; \frac{\partial r}{\partial z} = \frac{z}{r}]$$
$$= \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r}\frac{\partial}{\partial r}$$
$$= \frac{\vec{r}}{r}\frac{\partial}{\partial r}$$

Q# 57: Show that $\vec{\nabla}.(\phi\vec{A}) = \phi(\vec{\nabla}.\vec{A}) + (\vec{\nabla}\phi).\vec{A}$

Solution

Let $\vec{A} = A_1\hat{i} + A_2\hat{j} + A_3\hat{k}$ is a vector and ϕ is a function of a variable or variables

L.H.S $\vec{\nabla}.(\phi\vec{A}) = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}).(\phi\vec{A})$
$$= (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}).\phi(A_1\hat{i} + A_2\hat{j} + A_3\hat{k})$$
$$= (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}).(\phi A_1\hat{i} + \phi A_2\hat{j} + \phi A_3\hat{k})$$

$$= \frac{\delta}{\delta x}(\phi A_1) + \frac{\delta}{\delta y}(\phi A_2) + \frac{\delta}{\delta z}(\phi A_3) \qquad [\because \hat{i}.\hat{i} = 1; \hat{j}.\hat{j} = 1; \hat{k}.\hat{k} = 1]$$

$$= \phi\frac{\delta}{\delta x}(A_1) + A_1\frac{\delta}{\delta x}(\phi) + \phi\frac{\delta}{\delta y}(A_2) + A_2\frac{\delta}{\delta y}(\phi) + \phi\frac{\delta}{\delta z}(A_3) + A_3\frac{\delta}{\delta z}(\phi)$$

$$[\because \frac{d}{dx}(uv) = u\frac{d}{dx}v + v\frac{d}{dx}u]$$

$$= A_1\frac{\delta}{\delta x}(\phi) + A_2\frac{\delta}{\delta y}(\phi) + A_3\frac{\delta}{\delta z}(\phi) + \phi\frac{\delta}{\delta x}(A_1) + \phi\frac{\delta}{\delta y}(A_2) + \phi\frac{\delta}{\delta z}(A_3)$$

$$= A_1\frac{\delta}{\delta x}(\phi) + A_2\frac{\delta}{\delta y}(\phi) + A_3\frac{\delta}{\delta z}(\phi) + \phi\left\{\frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3)\right\}$$

$$= A_1\frac{\delta\phi}{\delta x} + A_2\frac{\delta\phi}{\delta y} + A_3\frac{\delta\phi}{\delta z} + \phi\left\{\frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3)\right\}$$

$$= \frac{\delta\phi}{\delta x}A_1 + \frac{\delta\phi}{\delta y}A_2 + \frac{\delta\phi}{\delta z}A_3 + \phi\left\{\frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3)\right\}$$

$$= \phi\left\{\frac{\delta}{\delta x}(A_1) + \frac{\delta}{\delta y}(A_2) + \frac{\delta}{\delta z}(A_3)\right\} + \frac{\delta\phi}{\delta x}A_1 + \frac{\delta\phi}{\delta y}A_2 + \frac{\delta\phi}{\delta z}A_3$$

$$= \phi(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}).(A_1\hat{i} + A_2\hat{j} + A_3\hat{k}) + (\frac{\delta\phi}{\delta x}\hat{i} + \frac{\delta\phi}{\delta y}\hat{j} + \frac{\delta\phi}{\delta z}\hat{k}).(A_1\hat{i} + A_2\hat{j} + A_3\hat{k})$$

$$[\because \hat{i}.\hat{i} = 1; \hat{j}.\hat{j} = 1; \hat{k}.\hat{k} = 1]$$

$$= \phi(\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}).(A_1\hat{i} + A_2\hat{j} + A_3\hat{k}) + (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k})\phi.(A_1\hat{i} + A_2\hat{j} + A_3\hat{k})$$

$$= \phi(\vec{\nabla}.\vec{A}) + (\vec{\nabla}\phi).\vec{A} \qquad [\because \vec{\nabla} = \frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k}]$$

$$\therefore \vec{\nabla}.(\phi\vec{A}) = \phi(\vec{\nabla}.\vec{A}) + (\vec{\nabla}\phi).\vec{A}$$

Q# 58: Show that $\nabla^2(\ln r) = \frac{1}{r^2}$

$$\text{L.H.S} = \nabla^2(\ln r)$$

$$= \vec{\nabla}.\vec{\nabla}(\ln r) \qquad [\because \vec{\nabla}.\vec{\nabla} = \nabla^2]$$

$$= \vec{\nabla}.\left[\frac{\vec{r}}{r}\frac{\partial}{\partial r}\ln r\right] \qquad [\because \vec{\nabla} = \frac{\vec{r}}{r}\frac{\partial}{\partial r}]$$

$$= \vec{\nabla}.\left[\frac{\vec{r}}{r}\frac{\partial}{\partial r}(\ln r)\right]$$

$$= \vec{\nabla}.\left[\frac{\vec{r}}{r}\frac{1}{r}\right] \qquad [\because \frac{\partial}{\partial r}(\ln r) = \frac{1}{r}]$$

$$= \vec{\nabla}.\left[\frac{\vec{r}}{r^2}\right]$$

$$= \vec{\nabla}.\left[\frac{1}{r^2}\vec{r}\right]$$

$$= \frac{1}{r^2}\left[\vec{\nabla}.\vec{r}\right] + \left[\vec{\nabla}\left(\frac{1}{r^2}\right)\right].\vec{r} \qquad [\because \vec{\nabla}.(\phi\vec{A}) = \phi(\vec{\nabla}.\vec{A}) + (\vec{\nabla}\phi).\vec{A}]$$

$$= \frac{3}{r^2} + \left[\frac{\vec{r}}{r}\frac{\partial}{\partial r}\left(\frac{1}{r^2}\right)\right].\vec{r} \qquad [\because \vec{\nabla}.\vec{r} = 3 \ \& \because \vec{\nabla} = \frac{\vec{r}}{r}\frac{\partial}{\partial r}]$$

$$= \frac{3}{r^2} + \left[\frac{\vec{r}}{r}\frac{\partial}{\partial r}\left(r^{-2}\right)\right].\vec{r}$$

$$= \frac{3}{r^2} + \frac{\vec{r}}{r}\left(-2r^{-2-1}\right).\vec{r}$$

$$= \frac{3}{r^2} + \frac{\vec{r}}{r}\left(-\frac{2}{r^3}\right).\vec{r}$$

$$= \frac{3}{r^2} - \frac{2}{r^4}\left(\vec{r}\,.\,\vec{r}\right)$$

$$= \frac{3}{r^2} - \frac{2}{r^4} \times r^2 \qquad [\because \vec{r}\,.\,\vec{r}\,. = r^2\,]$$

$$= \frac{3}{r^2} - \frac{2}{r^2}$$

$$= \frac{1}{r^2}.$$

Q# 59: Find the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r}$

Answer: Let $\phi = \frac{1}{r}$

Therefore, the directional derivative of $\frac{1}{r}$ in the direction of $\vec{r} = \vec{\nabla}\phi.\hat{r}$

We can write, $\hat{r} = \frac{\vec{r}}{\left|\vec{r}\right|} = \frac{\vec{r}}{r}$

Here, $\phi = \frac{1}{r}$

$$\Rightarrow \vec{\nabla}\phi = \vec{\nabla}\frac{1}{r}$$

$$\Rightarrow \vec{\nabla}\phi = \frac{\vec{r}}{r}\frac{\partial}{\partial r}\frac{1}{r} \qquad [\because \vec{\nabla} = \frac{\vec{r}}{r}\frac{\partial}{\partial r}]$$

$$= n\left[3r^{n-2} + \frac{\vec{r}}{r}\frac{\partial}{\partial r}(r^{n-2}).\vec{r}\right] \qquad [\because \vec{\nabla} = \frac{\vec{r}}{r}\frac{\partial}{\partial r}]\ \&\ [\vec{\nabla}\cdot\vec{r} = 3]$$

$$= n\left[3r^{n-2} + \frac{1}{r}\frac{\partial}{\partial r}(r^{n-2})(\vec{r}.\vec{r})\right]$$

$$= n\left[3r^{n-2} + \frac{1}{r}(n-2)(r^{n-2-1})(\vec{r}.\vec{r})\right]$$

$$= n\left[3r^{n-2} + \frac{1}{r}(n-2)r^{n-3}r^2\right] \qquad [\because \vec{r}.\vec{r} = r^2]$$

$$= n\left[3r^{n-2} + (n-2)r^{n-3}r\right]$$

$$= n\left[3r^{n-2} + (n-2)r^{n-3+1}\right]$$

$$= n\left[3r^{n-2} + (n-2)r^{n-2}\right]$$

$$= n[3 + n - 2]r^{n-2}$$

$$= n(n+1)r^{n-2}$$

Q# 61: Show that $\vec{\nabla}.(r^3\vec{r}) = 6r^3$

L.H.S $= \vec{\nabla}.(r^3\vec{r})$

$$= r^3(\vec{\nabla}.\vec{r}) + \vec{\nabla}(r^3).\vec{r} \qquad [\because \vec{\nabla}.(\phi\vec{A}) = \phi(\vec{\nabla}.\vec{A}) + (\vec{\nabla}\phi).\vec{A}]$$

$$= 3r^3 + \left[\frac{\vec{r}}{r}\frac{\partial}{\partial r}r^3\right]\cdot\vec{r} \qquad [\because \vec{\nabla}\cdot\vec{r} = 3\ \&\ \vec{\nabla} = \frac{\vec{r}}{r}\frac{\partial}{\partial r}]$$

$$= 3r^3 + \left[\frac{\vec{r}}{r}3r^{3-1}\right].\vec{r}$$

$$= 3r^3 + \frac{1}{r}3r^2(\vec{r}\cdot\vec{r})$$

$$= 3r^3 + 3r(r^2) \qquad [\because \vec{r}.\vec{r} = r^2]$$

$$= 3r^3 + 3r^3$$

$$= 6r^3$$

Q# 62: Show that $\vec{\nabla}.\left[r\vec{\nabla}\left(\frac{1}{r^3}\right)\right] = \frac{3}{r^4}$.

L.H.S $= \vec{\nabla}.\left[r\vec{\nabla}\left(\frac{1}{r^3}\right)\right]$

$$= \vec{\nabla}.\left[r\left\{\frac{\vec{r}}{r}\frac{\partial}{\partial r}\left(\frac{1}{r^3}\right)\right\}\right] \qquad [\because \vec{\nabla} = \frac{\vec{r}}{r}\frac{\partial}{\partial r}]$$

$$= \vec{\nabla}.\left[r\left\{\frac{\vec{r}}{r}\frac{\partial}{\partial r}r^{-3}\right\}\right]$$

Line Integral

Any Integral which is evaluated along the curve is called Line Integral, and it is denoted by $\int_C \vec{F}.d\vec{r}$ where $\vec{F}$ is a vector point function, $\vec{r}$ is a position vector and C is the curve

Theorem: the work performed by a vector field on a particle moving along a parametric curve C is obtained by integrating the scalar tangential component of force along C.

$$W = \oint_C \vec{F}.\vec{dr}$$

Q # 74: Establish the path Integral $\int_C \vec{F}.d\vec{S} = \lim_{n\to\infty} \sum_{i=1}^{n} \vec{F}(\vec{S}_{i-1}).(\vec{S}_i - \vec{S}_{i-1})$

Answer:

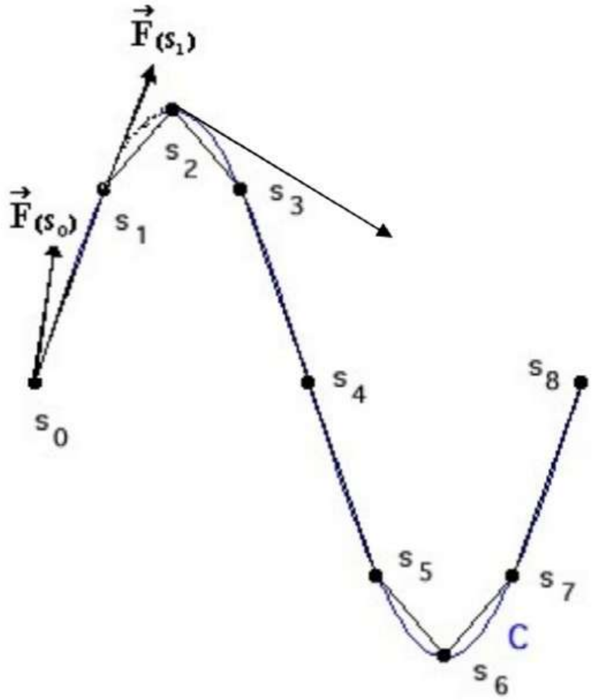

Figure # 82

We choose points $S_0, S_1, S_2, \ldots\ldots\ldots\ldots\ldots S_n$ along the path **C** and then connect these points as shown in the figure above.

and the amount of total work done on the whole path by :

$$W_1 = \vec{F}(S_0).(\vec{S}_1 - \vec{S}_0) = \vec{F}(S_{1-1}).(\vec{S}_1 - \vec{S}_{1-1})$$

$$W_2 = \vec{F}(S_1).(\vec{S}_2 - \vec{S}_1) = \vec{F}(S_{2-1}).(\vec{S}_2 - \vec{S}_{2-1})$$

$$W_3 = \vec{F}(S_2).(\vec{S}_3 - \vec{S}_2) = \vec{F}(S_{3-1}).(\vec{S}_3 - \vec{S}_{3-1})$$

$$\mathbf{W_n = \vec{F}_{(S_{n-1})} . (\vec{S}_n - \vec{S}_{n-1})}$$

Total Work: $\mathbf{W_1 + W_2 + W_3 + --------- + W_n = \sum_{i=1}^{n} W_i = \sum_{i=1}^{n} \vec{F}_{(\vec{S}_{i-1})} . (\vec{S}_i - \vec{S}_{i-1})}$

Then we estimate the Total work done on the **i**-th segment of the path by

$$\mathbf{\sum_{i=1}^{n} W_i = \vec{F}_{(S_{i-1})} . (\vec{S}_i - \vec{S}_{i-1})}$$

By using a large number of small segments we can obtain a very good estimate for the amount of work done. The exact amount of work done is obtained by taking the limit of these estimates. This limit is called the **line integral** of the vector field **F** over the path **C** and the amount of work done on the whole path by:

Total Work done = $\mathbf{\int_C \vec{F} . d\vec{S} = \lim_{n \to \infty} \sum_{i=1}^{n} \vec{F}_{(\vec{S}_{i-1})} . (\vec{S}_i - \vec{S}_{i-1})}$ ***Answer***

[Where $ds = s_1 - s_0 = s_2 - s_1 = s_3 - s_2 = \ldots\ldots\ldots\ldots\ldots = s_n - s_{n-1}$]

Mathematical Expression to find out work done along a curve

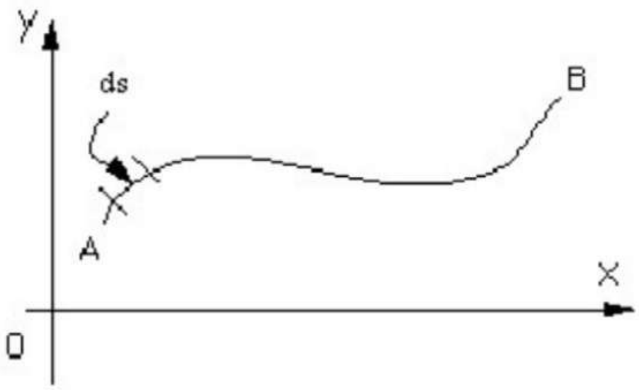

Figure # 83

Then the work done by that force in moving the particle small distance ds is given by: Work Done $= \mathbf{F \times ds}$

Now if we want to know the total work done in moving the particle all the way from A to B, we need to add up all the small contributions, each of the form $\mathbf{F \times ds}$.

However $\mathbf{F}$ may have different strengths at different positions, i.e. $\mathbf{F}$ is a function of position, so what we need to add up are lots of contributions like $\mathbf{F(s) \times ds}$.

Answer:

Figure shows the path of the particle, Call this path OP. Then

Here, $\vec{\mathbf{F}}(\mathbf{x},\mathbf{y}) = \mathbf{xy}\,\hat{\mathbf{i}} + \mathbf{y}\,\hat{\mathbf{j}}$,---------------------------------(i)

We know,

$\vec{\mathbf{F}}(\mathbf{x},\mathbf{y}) = \mathbf{P}\,\hat{\mathbf{i}} + \mathbf{Q}\,\hat{\mathbf{j}}$---(ii)

Comparing (i) and (ii),

$\mathbf{p = xy, Q = y}$

We know,

Work = $\int_{OP} \vec{\mathbf{F}}.d\vec{\mathbf{S}} = \int_{OP} (\mathbf{Pdx + Qdy})$

Work = $\int_{OP} (\mathbf{xydx + ydy})$ $[\because \mathbf{p = xy, Q = y}]$

To evaluate this integral, let us use x as the parameter,

Then, Given $\mathbf{y = x^2}$

$$\frac{dy}{dx} = \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = 2x$$

$$\therefore dy = 2xdx$$

$$\text{Work} = \int_o^p (xydx + ydy)$$

$$\text{Work} = \int_0^3 \left[(x.x^2dx + x^2.2xdx\right]$$

$$\text{Work} = \int_0^3 (x^3dx + 2x^3dx)$$

$$\text{Work} = \int_0^3 3x^3dx = 3\left[\frac{x^{3+1}}{3+1}\right]_0^3 = \frac{3}{4}\left[x^4\right]_0^3 = \frac{3}{4} \times 3^4 = \frac{243}{4} \textit{ Answer.}$$

Q # 76: Evaluate $\int_C \mathbf{xy\,dx}$ from **B(1,0)** to **C(0,1)** along the curve C that is the portion of $\mathbf{x^2 + y^2 = 1}$ in the first quadrant.

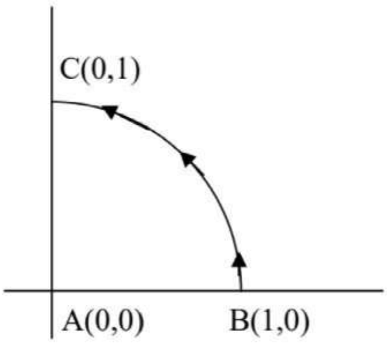

Figure # 88

Anwer:

Given $\mathbf{x^2 + y^2 = 1}$

$\Rightarrow \mathbf{y = \sqrt{1 - x^2}}$

The curve BC is the first quadrant of the unit circle as shown in figure. On the curve $\mathbf{y = \sqrt{1 - x^2}}$, so that,

$$\int_C xy\,dx = \int_1^0 x\sqrt{1 - x^2}\,dx \qquad \text{Let } 1 - x^2 = z$$

$$-2x\,dx = dz$$

$$x\,dx = -\frac{dz}{2}$$

x	1	0
$1 - x^2 = z$ $\therefore z = 1 - x^2$	$z = 1 - x^2$ $z = 1 - 1$ $z = 0$	$z = 1 - x^2$ $z = 1 - 0$ $z = 1$

$$\int_C xy\,dx = \int_1^0 x\sqrt{1 - x^2}\,dx =$$

$$-\int_0^1 \sqrt{z}\,\frac{dz}{2} = -\frac{1}{2}\int_0^1 \sqrt{z}\,dz = -\frac{1}{2}\left[\frac{z^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^1 = -\frac{1}{2}\left[\frac{1^{\frac{3}{2}}}{\frac{3}{2}} - \frac{0^{\frac{3}{2}}}{\frac{3}{2}}\right] = -\frac{1}{2}\left[\frac{1}{\frac{3}{2}} - \frac{0}{\frac{3}{2}}\right]$$

$$= -\frac{1}{2} \times \frac{2}{3}(1 - 0) = -\frac{1}{3} \quad \textit{Answer}$$

Q # 77: Find the value of the line integral when $\vec{\mathbf{F}}(\mathbf{r}) = -y\,\hat{i} - xy\,\hat{j}$, where $\vec{\mathbf{r}}$ is a function of $\mathbf{t}$ and $\mathbf{C}$ is the circular arc in Figure from A to B.

Or

Find the work done in moving a particle once around a quarter circle C in the xy plane, if the circle has center at the origin and radius 1 and if the force field is given by $\vec{\mathbf{F}}(\mathbf{r}) = -y\,\hat{i} - xy\,\hat{j}$

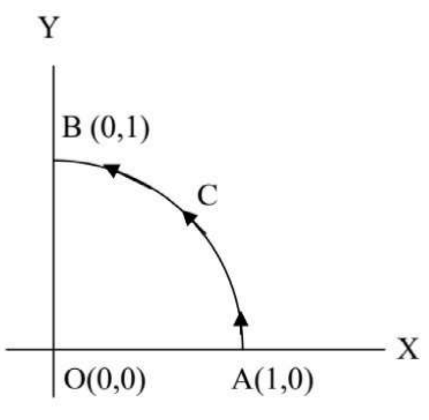

Figure # 89

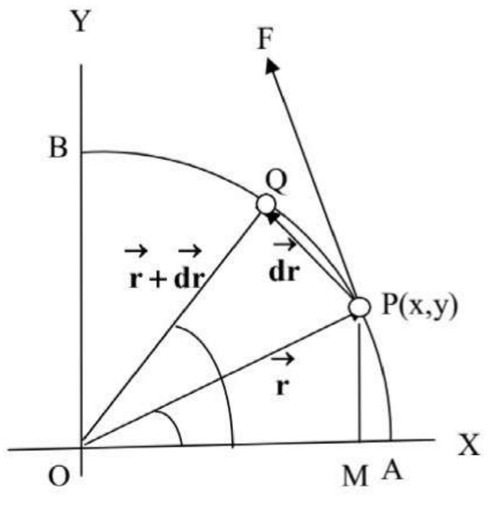

Figure # 90

$$W = \left[\frac{-63}{6}\right] + \left[\frac{18}{3}\right] + \left[\frac{30}{4}\right]$$

$$W = \frac{-126 + 72 + 90}{12}$$

$$W = \frac{-126 + 162}{12}$$

$W = \frac{36}{12}$=3; where the units for W depend on the units chosen for force and distance

Q # 80: Find the value of $\int_C \vec{F}.\vec{dr}$ where $\vec{F} = (y - 2x)\hat{i} + (3x + 2y)\hat{j}$ and C is a circle in the xy- plane with center the origin and radius 2.

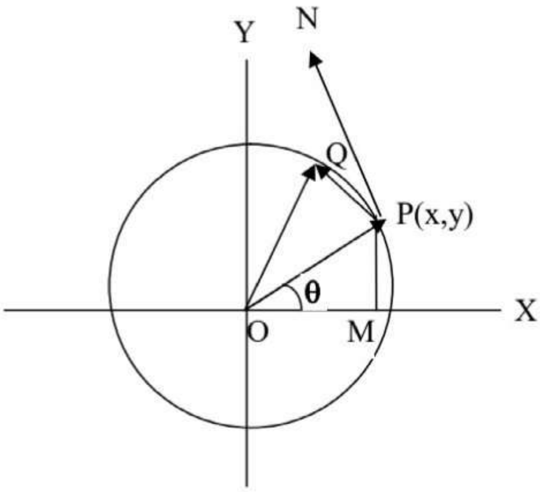

Figure # 94

The position Vector is

$\vec{r} = x\hat{i} + y\hat{j}$--------------------------(i) **[Page no 48, Figure no 57, Equation no (i)]**

Let,

$\angle POM = \theta$, $OP = 2$

$$\frac{PM}{OP} = \sin\theta$$

$$\frac{y}{2} = \sin\theta \Rightarrow y = 2\sin\theta$$

Similarly,

$$\frac{OM}{OP} = \cos\theta \Rightarrow \frac{x}{2} = \cos\theta \Rightarrow x = 2\cos\theta$$

From (i),

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$\vec{r}(\theta) = 2\cos\theta\,\hat{i} + 2\sin\theta\,\hat{j}$$

$$\frac{d\vec{r}}{d\theta} = -2\sin\theta\,\hat{i} + 2\cos\theta\,\hat{j}$$

$$\Rightarrow d\vec{r} = (-2\sin\theta\,\hat{i} + 2\cos\theta\,\hat{j})d\theta$$

Given,

$$\vec{F} = (y-2x)\,\hat{i} + (3x+2y)\,\hat{j}$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int \{(y-2x)\hat{i} + (3x+2y)\hat{j}\}.\ (-2\sin\theta\,\hat{i} + 2\cos\theta\,\hat{j})d\theta$$

$[\because \hat{i}.\hat{i} = 1,\ \hat{j}.\hat{j} = 1,\ \hat{k}.\hat{k} = 1,\ \hat{i}.\hat{j} = 0\ ,\ \hat{i}.\hat{k} = 0\ ,\ \hat{j}.\hat{i} = 0\ ,\ \hat{j}.\hat{k} = 0\ ,\ \hat{k}.\hat{i} = 0\ ,\ \hat{k}.\hat{j} = 0\ ]]$

$$\therefore \int_C \vec{F}.\vec{dr} = \int \{(y-2x)(-2\sin\theta)d\theta + (3x+2y)(2\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int \{(2\sin\theta - 2\times 2\cos\theta)(-2\sin\theta)d\theta + (3\times 2\cos\theta + 2\times 2\sin\theta)(2\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int \{(2\sin\theta - 4\cos\theta)(-2\sin\theta)d\theta + (6\cos\theta + 4\sin\theta)(2\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int \{(-4\sin^2\theta + 8\sin\theta\cos\theta)d\theta + (12\cos^2\theta + 8\sin\theta\cos\theta)d\theta\}.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int (-4\sin^2\theta + 16\sin\theta\cos\theta + 12\cos^2\theta)d\theta.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int_0^{2\pi} (-4\sin^2\theta + 16\sin\theta\cos\theta + 12\cos^2\theta)d\theta.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int_0^{2\pi} (-2\times 2\sin^2\theta + 8\times 2\sin\theta\cos\theta + 6\times 2\cos^2\theta)d\theta.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \int_0^{2\pi} \{-2(1-\cos 2\theta) + 8\times\sin 2\theta + 6(1+\cos 2\theta)\}d\theta.$$

$$\therefore \int_C \vec{F}.\vec{dr} = -2\int_0^{2\pi} d\theta + 2\int_0^{2\pi}\cos 2\theta\, d\theta + 8\times\int_0^{2\pi}\sin 2\theta\, d\theta + 6\int_0^{2\pi} d\theta + 6\int_0^{2\pi}\cos 2\theta\, d\theta.$$

$$\therefore \int_C \vec{F}.\vec{dr} = \left[-2\theta + 2\frac{\sin 2\theta}{2} - 8\times\frac{\cos 2\theta}{2} + 6\theta + 6\frac{\sin 2\theta}{2}\right]_0^{2\pi}$$

$$\therefore \int_C \vec{F}.\vec{dr} = \left[-2(\theta - \frac{\sin 2\theta}{2}) - 8\times\frac{\cos 2\theta}{2} + 6(\theta + \frac{\sin 2\theta}{2})\}\right]_0^{2\pi}$$

$$\therefore \int_C \vec{F}.\vec{dr}$$

$$= -2\left[\left(2\pi - \frac{\sin 2\times 2\pi}{2}\right) - \left(0 - \frac{\sin 2\times 0}{2}\right)\right] - 8\left[\left(\frac{\cos 2\times 2\pi}{2} - \frac{\cos 2\times 0}{2}\right)\right] + 6\left[\left(2\pi + \frac{\sin 2\times 2\pi}{2}\right) - \left(0 + \frac{\sin 2\times 0}{2}\right)\right]$$

$$= -2\left[\left(2\pi - \frac{\sin 4\pi}{2}\right) - \left(0 - \frac{\sin 0}{2}\right)\right] - 8\left[\left(\frac{\cos 4\pi}{2} - \frac{\cos 0}{2}\right)\right] + 6\left[\left(2\pi + \frac{\sin 4\pi}{2}\right) - \left(0 + \frac{\sin 0}{2}\right)\right]$$

$$= -2\left[\left(2\pi - \frac{0}{2}\right) - \left(0 - \frac{0}{2}\right)\right] - 8\left[\left(\frac{1}{2} - \frac{1}{2}\right)\right] + 6\left[\left(2\pi + \frac{0}{2}\right) - \left(0 + \frac{0}{2}\right)\right]$$

$$= -2\left[2\pi - \left(-\frac{0}{2}\right)\right] - 8[0] + 6[2\pi - 0]$$

$$= -2[2\pi + 0] - 0 + 6[2\pi]$$

$$= -2[2\pi] + 6[2\pi]$$

$$\therefore \int_C \vec{F}.\vec{dr}\ [= -4\pi + 12\pi = 8\pi \text{ Answer}$$

Q # 81: Find the value of $\int_C \vec{F}.\vec{dr}$ where $\vec{F} = x^2\,\hat{i} + 3xy\,\hat{j}$ if

a) C is the straight line path from $(0,0)$ to $(1,2)$

b) C is the parabolic path $y = 2x^2$ from $(0,0)$ to $(1,2)$

c) C is composed of two straight-line paths the x axis from $(0,0)$ to $(1,0)$ and then a line parallel to the y-axis from $(1,0)$ to $(1,2)$

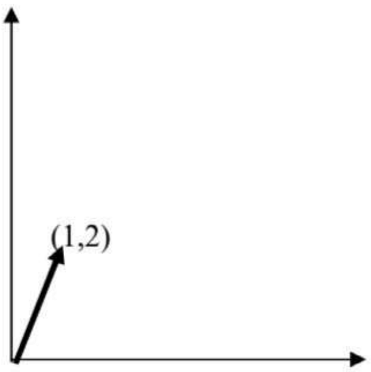

Figure # 95

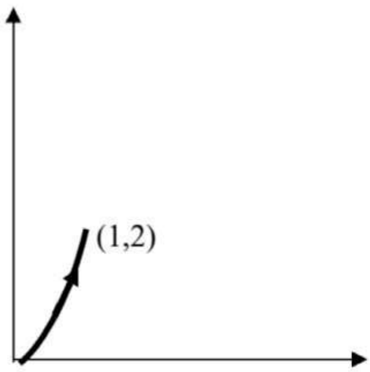

Figure # 96

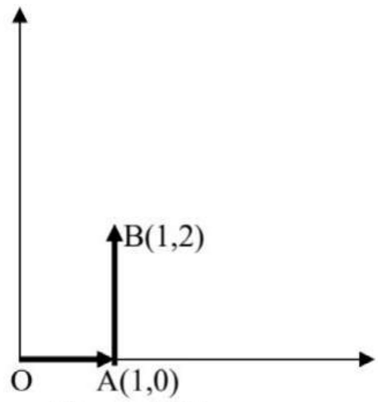

Figure # 97

Let, the position vector is: $\vec{r} = x\,\hat{i} + y\,\hat{j}$ **[Page no 48, Figure no 57, Equation no (i)]**

Since the line element $\vec{dr}$ lies in the xy-plane, we can express it as $\vec{dr} = dx\,\hat{i} + dy\,\hat{j}$, so that

$$\int_C \vec{F}.\vec{dr} = \int_C (x^2\,\hat{i} + 3xy\,\hat{j}).(dx\,\hat{i} + dy\,\hat{j})$$

$$= \int_C (x^2dx + 3xydy) \text{ -------------------------------------(i)}$$

a) On this path $y = 2x$, so we can convert (i) to a definite integral with respect to x, this is effectively a simple parameterization, $x = t, y = 2t, 0 \le t \le 1$

Hence,

$$\oint_{C_3}\left[y^2dx - x^2dy\right] = \int_2^0\left[x^4dx - x^2(2xdx)\right] = \int_2^0 (x^4dx - 2x^3)dx$$

$$= \left[\frac{1}{5}x^5 - \frac{1}{2}x^4\right]_2^0 = \frac{8}{5}$$

Therefore, $\oint_C y^2dx - x^2dy = 0 - 16 + \frac{8}{5} = -\frac{72}{5}$

Q # 84: Evaluate the line integral, $\int \vec{F}\,.\vec{dr}$ where the force field is given by $\vec{F}(x,y) = 3xy\,\hat{i} - 5z\,\hat{j} + 10x\,\hat{k}$ along the curve, $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$

Answer:

We have the position vector

$\therefore \vec{r} = x\,\hat{i} + y\,\hat{j} + z\,\hat{k}$ **[Page no 48, Figure no 57, Equation no (i)]**

$\therefore \vec{dr} = dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$

$\vec{F}\,.\vec{dr} = [3xy\,\hat{i} - 5z\,\hat{j} + 10x\,\hat{k}].\,[dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}]$

$\vec{F}\,.\vec{dr} = 3xydx - 5zdy + 10xdz$--------------------------(i)

Given

$x = t^2 + 1$

$\therefore \frac{dx}{dt} = \frac{d}{dt}(t^2 + 1)$

$\therefore \frac{dx}{dt} = 2t$

$\therefore dx = 2tdt$

$y = 2t^2$

$\therefore \frac{dy}{dt} = \frac{d}{dt}(2t^2)$

$\therefore \frac{dy}{dt} = 4t$

$\therefore dy = 4tdt$

$z = t^3$

$\therefore \frac{dz}{dt} = \frac{d}{dt}(t^3)$

$\therefore \dfrac{dz}{dt} = 3t^2$

$\therefore dz = 3t^2 dt$

From (i),

$\vec{F}.\vec{dr} = 3xydx - 5zdy + 10xdz$

$\vec{F}.\vec{dr} = 3(t^2+1)(2t^2)2tdt - 5t^3 4tdt + 10(t^2+1)3t^2dt$

$\vec{F}.\vec{dr} = 3(t^2+1)(4t^3)dt - 20t^4dt + 10(t^2+1)3t^2dt$

$\vec{F}.\vec{dr} = 3(4t^5+4t^3)dt - 20t^4dt + 10(3t^4+3t^2)dt$

$\vec{F}.\vec{dr} = (12t^5+12t^3)dt - 20t^4dt + (30t^4+30t^2)dt$

$\vec{F}.\vec{dr} = 12t^5dt + 12t^3dt - 20t^4dt + 30t^4dt + 30t^2dt$

$\vec{F}.\vec{dr} = 12t^5dt + 10t^4dt + 12t^3dt + 30t^2dt$

Total work done$= \int_1^2 \vec{F}.\vec{dr} = \int_1^2 (12t^5dt + 10t^4dt + 12t^3dt + 30t^2dt)$

$$= \int_1^2 12t^5dt + \int_1^2 10t^4dt + \int_1^2 12t^3dt + \int_1^2 30t^2dt$$

$$= \left[12\frac{t^6}{6} + 10\frac{t^5}{5} + 12\frac{t^4}{4} + 30\frac{t^3}{3}\right]_1^2$$

$$= [2t^6 + 2t^5 + 3t^4 + 10t^3]_1^2$$

$$= (2\times2^6 + 2\times2^5 + 3\times2^4 + 10\times2^3) - (2\times1^6 + 2\times1^5 + 3\times1^4 + 10\times1^3)$$

$$= (2\times64 + 2\times32 + 3\times16 + 10\times8) - (2+2+3+10)$$

$$= (128+64+48+80) - (2+2+3+10)$$

$$= (320) - (17)$$

$$= 303$$

Q # 85: If $\vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$ and C is the curve curve, $x = t^2$, $y = 2t$, $z = t^3$ from $t = 0$ to $t = 1$, then evaluate the line integral, $\int_c \vec{F} \times \vec{dr}$

Answer: We have the position vector

$\therefore \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ **[Page no 48, Figure no 57, Equation no (i)]**

$$= \left[9z\hat{i} + 6\frac{z^2}{2}\hat{j} - 8z\hat{k}\right]_0^4 \qquad [\because \int x^n dx = \frac{x^{n+1}}{n+1}; \int dz = z]$$

$$= \left[9z\hat{i} + 3z^2\hat{j} - 8z\hat{k}\right]_0^4$$

$$= \left[(9\times 4\hat{i} + 3\times 4^2\hat{j} - 8\times 4\hat{k}) - (9\times 0\hat{i} + 3\times 0^2\hat{j} - 8\times 0\hat{k})\right]$$

$$= \left[36\hat{i} + 3\times 16\hat{j} - 32\hat{k}\right]$$

$$= 4\left[9\hat{i} + 12\hat{j} - 8\hat{k}\right]$$

Q # 90: Show that $\iint \vec{F}.\hat{\eta} ds = \frac{3}{2}$; Where $\vec{F} = 4xz\,\hat{i} - y^2\,\hat{j} + yz\,\hat{k}$ and S is the surface of the cube bounded by the planes $\mathbf{x = 0, x = 1, y = 0, y = 1, z = 0, z = 1}$

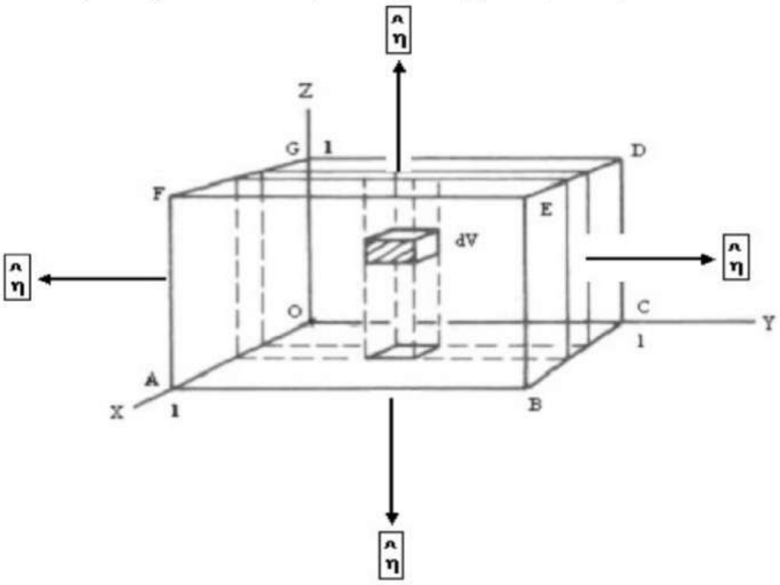

Figure # 113

Serial no	Surface	$\hat{\eta}$ =?	ds	Plane
1	OABC	$\hat{\eta} = -\hat{k}$	**dxdy**	$\mathbf{z = 0}$
2	DEFG	$\hat{\eta} = \hat{k}$	**dxdy**	$\mathbf{z = 1}$
3	OAFG	$\hat{\eta} = -\hat{j}$	**dxdz**	$\mathbf{y = 0}$
4	BCDE	$\hat{\eta} = \hat{j}$	**dxdz**	$\mathbf{y = 1}$
5	OCDG	$\hat{\eta} = -\hat{i}$	**dydz**	$\mathbf{x = 0}$
6	ABEF	$\hat{\eta} = \hat{i}$	**dydz**	$\mathbf{x = 1}$

Now,

$$\iint_S \vec{F}.\hat{\eta}ds = \iint_{OABC}\vec{F}.\hat{\eta}ds + \iint_{DEFG}\vec{F}.\hat{\eta}ds + \iint_{OAFG}\vec{F}.\hat{\eta}ds + \iint_{BCDE}\vec{F}.\hat{\eta}ds + \iint_{OCDG}\vec{F}.\hat{\eta}ds + \iint_{ABEF}\vec{F}.\hat{\eta}ds \text{ -(1)}$$

1. $$\iint_{OABC}\vec{F}.\hat{\eta}ds = \iint_S (4xz\hat{i} - y^2\hat{j} + yz\hat{k}).\hat{\eta}ds = \iint_S (4xz\hat{i} - y^2\hat{j} + yz\hat{k}).(-\hat{k})dxdy$$

$[\because \hat{i}.\hat{i} = 1, \hat{j}.\hat{j} = 1, \hat{k}.\hat{k} = 1, \hat{i}.\hat{j} = 0, \hat{i}.\hat{k} = 0, \hat{j}.\hat{i} = 0, \hat{j}.\hat{k} = 0, \hat{k}.\hat{i} = 0, \hat{k}.\hat{j} = 0\ ]$

$$= \int_0^1\int_0^1 -yzdxdy$$

$$= -\int_0^1\int_0^1 yzdxdy = -\int_0^1 [yzx]_0^1 dy \qquad [\because \int dx = x]$$

$$= -\int_0^1 [yz \times 1 - yz \times 0]dy = -\int_0^1 [yz(1-0)]dy$$

$$= -\left[z\frac{y^2}{2}\right]_0^1 = -\left[z\frac{1^2}{2} - z\frac{0^2}{2}\right] = -\frac{z}{2}(1-0) = -\frac{z}{2} = 0 [\because z = 0]$$

2.

$$\iint_{DEFG}\vec{F}.\hat{\eta}ds = \iint_S (4xz\hat{i} - y^2\hat{j} + yz\hat{k}).\hat{\eta}ds = \iint_S (4xz\hat{i} - y^2\hat{j} + yz\hat{k}).(\hat{k})dxdy = \int_0^1\int_0^1 yzdxdy$$

$[\because \hat{i}.\hat{i} = 1, \hat{j}.\hat{j} = 1, \hat{k}.\hat{k} = 1, \hat{i}.\hat{j} = 0, \hat{i}.\hat{k} = 0, \hat{j}.\hat{i} = 0, \hat{j}.\hat{k} = 0, \hat{k}.\hat{i} = 0, \hat{k}.\hat{j} = 0\ ]$

$$= \int_0^1\int_0^1 yzdxdy = \int_0^1 [yzx]_0^1 dy = \int_0^1 [yz \times 1 - yz \times 0]dy \qquad [\because \int dx = x]$$

$$= \int_0^1 [yz]dy = \left[z\frac{y^2}{2}\right]_0^1 = \left[z\frac{1^2}{2} - z\frac{0^2}{2}\right] = \frac{z}{2}(1-0) = \frac{z}{2} = \frac{1}{2} [\because z = 1]$$

3.

$$\iint_{OAFG}\vec{F}.\hat{\eta}ds = \iint_S (4xz\hat{i} - y^2\hat{j} + yz\hat{k}).\hat{\eta}ds = \iint_S (4xz\hat{i} - y^2\hat{j} + yz\hat{k}).(-\hat{j})dxdz = \int_0^1\int_0^1 y^2dxdz$$

$[\because \hat{i}.\hat{i} = 1, \hat{j}.\hat{j} = 1, \hat{k}.\hat{k} = 1, \hat{i}.\hat{j} = 0, \hat{i}.\hat{k} = 0, \hat{j}.\hat{i} = 0, \hat{j}.\hat{k} = 0, \hat{k}.\hat{i} = 0, \hat{k}.\hat{j} = 0\ ]$

$$= \int_0^1\int_0^1 y^2dxdz = \int_0^1 [xy^2]_0^1 dz \qquad [\because \int dx = x]$$

$$= \int_0^1 [1 \times y^2 - 0 \times y^2]dz$$

$$= \int_0^1 [y^2 - 0]dz = [y^2z]_0^1 = [y^2 \times 1 - y^2 \times 0] \qquad [\because \int dz = z]$$

$$= y^2(1-0) = y^2 = 0 \qquad [\because y = 0]$$

4.

$$\iint_{BCDE}\vec{F}.\hat{\eta}ds=\iint_S(4xz\hat{i}-y^2\hat{j}+yz\hat{k}).\hat{\eta}ds=\iint_S(4xz\hat{i}-y^2\hat{j}+yz\hat{k}).(\hat{j})dxdz=\int_0^1\int_0^1-y^2dxdz$$

$[\because \hat{i}.\hat{i}=1,\ \hat{j}.\hat{j}=1,\hat{k}.\hat{k}=1,\hat{i}.\hat{j}=0\ ,\hat{i}.\hat{k}=0\ ,\hat{j}.\hat{i}=0\ ,\hat{j}.\hat{k}=0\ ,\hat{k}.\hat{i}=0\ ,\hat{k}.\hat{j}=0\ ]$

$$=-\int_0^1\int_0^1 y^2dxdz=-\int_0^1\left[xy^2\right]_0^1dz=-\int_0^1\left[1\times y^2-0\times y^2\right]dz \qquad [\because \int dx=x]$$

$$=-\int_0^1\left[y^2-0\right]dz=-\left[y^2z\right]_0^1 \qquad [\because \int dz=z]$$

$$=-\left[y^2\times 1-y^2\times 0\right]$$

$$=-y^2(1-0)=-y^2=-1 \qquad [\because y=1]$$

5. $$\iint_{OCDG}\vec{F}.\hat{\eta}ds=\iint_S(4xz\hat{i}-y^2\hat{j}+yz\hat{k}).\hat{\eta}ds=\iint_S(4xz\hat{i}-y^2\hat{j}+yz\hat{k}).(-\hat{i})dydz$$

$[\because \hat{i}.\hat{i}=1,\ \hat{j}.\hat{j}=1,\hat{k}.\hat{k}=1,\hat{i}.\hat{j}=0\ ,\hat{i}.\hat{k}=0\ ,\hat{j}.\hat{i}=0\ ,\hat{j}.\hat{k}=0\ ,\hat{k}.\hat{i}=0\ ,\hat{k}.\hat{j}=0\ ]$

$$=\int_0^1\int_0^1-4xzdydz$$

$$=-\int_0^1\int_0^1 4xzdydz=-\int_0^1\left[4xyz\right]_0^1dz=-\int_0^1\left[4x\times 1\times z-4x\times 0\times z\right]dz \qquad [\because \int dy=y]$$

$$=-\int_0^1\left[4xz(1-0)\right]dz=-\left[4x\frac{z^2}{2}\right]_0^1=-\left[4x\frac{1^2}{2}-4x\frac{0^2}{2}\right]$$

$$=-4x(\frac{1}{2}-0)=-2x=0 \qquad [\because x=0]$$

6. $$\iint_{ABEF}\vec{F}.\hat{\eta}ds=\iint_S(4xz\hat{i}-y^2\hat{j}+yz\hat{k}).\hat{\eta}ds=\iint_S(4xz\hat{i}-y^2\hat{j}+yz\hat{k}).(\hat{i})dydz$$

$[\because \hat{i}.\hat{i}=1,\ \hat{j}.\hat{j}=1,\hat{k}.\hat{k}=1,\hat{i}.\hat{j}=0\ ,\hat{i}.\hat{k}=0\ ,\hat{j}.\hat{i}=0\ ,\hat{j}.\hat{k}=0\ ,\hat{k}.\hat{i}=0\ ,\hat{k}.\hat{j}=0\ ]$

$$=\int_0^1\int_0^1 4xzdydz$$

$$=\int_0^1\int_0^1 4xzdydz=\int_0^1\left[4xyz\right]_0^1dz=\int_0^1\left[4x\times 1\times z-4x\times 0\times z\right]dz \qquad [\because \int dy=y]$$

$$=\int_0^1\left[4xz(1-0)\right]dz=\left[4x\frac{z^2}{2}\right]_0^1=\left[4x\frac{1^2}{2}-4x\frac{0^2}{2}\right]$$

$$=4x(\frac{1}{2}-0)=2x=2 \qquad [\because x=1]$$

Putting the values in (1),

Now,

$$\iint_{S} \vec{F}.\hat{\eta}ds = \iint_{OABC} \vec{F}.\hat{\eta}ds + \iint_{DEFG} \vec{F}.\hat{\eta}ds + \iint_{OAFG} \vec{F}.\hat{\eta}ds + \iint_{BCDE} \vec{F}.\hat{\eta}ds + \iint_{OCDG} \vec{F}.\hat{\eta}ds + \iint_{ABEF} \vec{F}.\hat{\eta}ds$$

$$\iint_{S} \vec{F}.\hat{\eta}ds = 0 + \frac{1}{2} + 0 - 1 + 0 + 2 = \frac{1}{2} + 1 = \frac{3}{2} \text{ (Proved)}$$

Area Enclosed by a closed Curve:

One of the earliest applications of integration is finding the area of a plane figure bounded by the x-axis, the curve $y = f(x)$ and ordinates at $x = x_1$ and $x = x_2$.

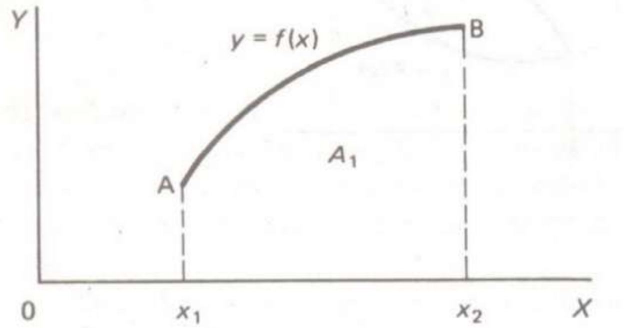

Figure # 114

$$A_1 = \int_{x_1}^{x_2} ydx = \int_{x_1}^{x_2} f(x)dx \text{ ------------------------------------(i)}$$

If points A and B are joined by another curve $y = F(x)$

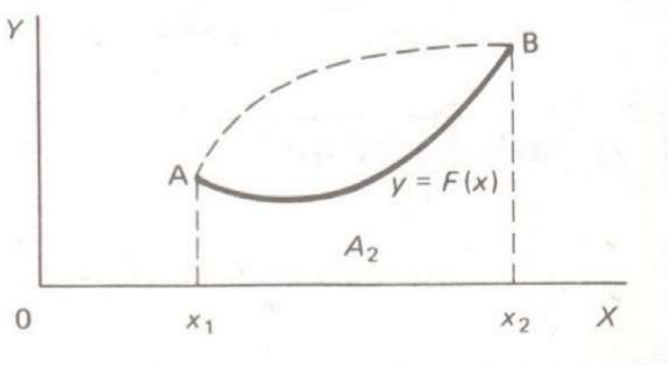

Figure # 115

$$A_2 = \int_{x_1}^{x_2} ydx = \int_{x_1}^{x_2} F(x)dx \text{ ----------------------------------(ii)}$$

Combining the two figures, we have

Q # 91: Determine the area enclosed by the graphs of $\mathbf{y = x^3}$ and $\mathbf{y = 4x}$ for $\mathbf{x \geq 0}$

Answer: First we need to know the points of intersection.

Given, $\mathbf{y = x^3}$ and $\mathbf{y = 4x}$

Then, we can write, $\mathbf{x^3 = 4x}$

$$\Rightarrow x^3 - 4x = 0$$
$$\Rightarrow x(x^2 - 4) = 0$$
$$\Rightarrow x(x+2)(x-2) = 0$$

$\Rightarrow \mathbf{x = 0}$ and $\mathbf{x = -2}$, $\mathbf{x = 2}$

But $\mathbf{x \geq 0}$

$\therefore \mathbf{x = 0}$ and $\mathbf{x = 2}$

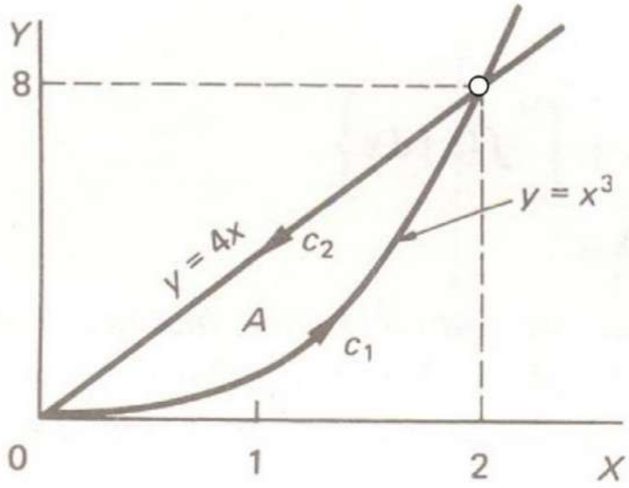

Figure # 119

We integrate in an anticlockwise manner

$\mathbf{c_1 : y = x^3}$, Limits $\mathbf{x = 0}$ to $\mathbf{x = 2}$

$\mathbf{c_2 : y = 4x}$, Limits $\mathbf{x = 2}$ to $\mathbf{x = 0}$

We have,

$$\therefore A = -\oint y\,dx = -\{\int_{x_1}^{x_2} F(x)dx + \int_{x_2}^{x_1} f(x)dx\}$$
$$= -\{\int_0^2 F(x)dx + \int_2^0 f(x)dx\}$$
$$= -\{\int_0^2 x^3 dx + \int_2^0 4x\,dx\}$$
$$= -\{\left[\frac{x^4}{4}\right]_0^2 + 4\left[\frac{x^2}{2}\right]_2^0\}$$

$$= -\{\left[\frac{2^4}{4} - 0\right] + 4\left[\frac{0^2}{2} - \frac{2^2}{2}\right]\}$$

$$= -\{\left[\frac{16}{4}\right] + 4\left[-\frac{4}{2}\right]\}$$

$= \mathbf{4}$ Answer

Q # 92: Find the area of the triangle with vertices **(0,0)** ,**(5,3)** and **(2,6)**

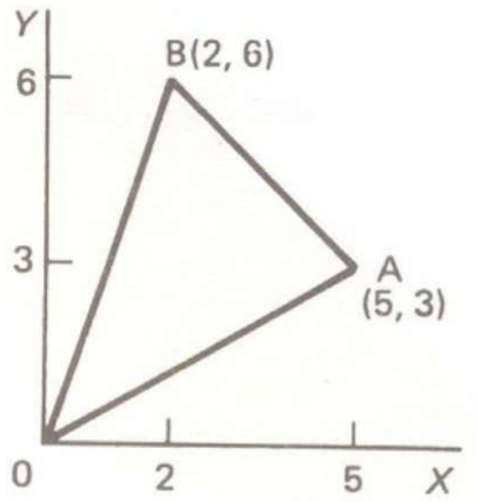

Figure # 120

Answer: We have,

$$\frac{y - y_1}{y_1 - y_2} = \frac{x - x_1}{x_1 - x_2}$$

$\therefore$ The equation of **OA** or **AO** is $\frac{y-0}{0-3} = \frac{x-0}{0-5}$

$$\Rightarrow \frac{y}{-3} = \frac{x}{-5}$$

$$\Rightarrow \frac{y}{3} = \frac{x}{5}$$

$$\Rightarrow y = \frac{3x}{5}$$

The equation of **BA** or **AB** is $\frac{y-6}{6-3} = \frac{x-2}{2-5}$

$$\Rightarrow \frac{y-6}{3} = \frac{x-2}{-3}$$

$$\Rightarrow \frac{y-6}{1} = \frac{x-2}{-1}$$

$$\Rightarrow \frac{y-6}{1} = -x + 2$$

$\mathbf{OA : c_1}$ is the line $\mathbf{y = 0 \therefore dy = 0}$, Then

$\mathbf{I_1 = \oint(x^2dx - 2xydy)}$ for this part becomes

$$\mathbf{I_1 = \int_0^1 (x^2dx - 2x.0.0) = \int_0^1 x^2dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}}$$

b) $\mathbf{AB : c_2}$ is the line: $\mathbf{\frac{y-0}{0-1} = \frac{x-1}{1-0} \Rightarrow \frac{y}{-1} = \frac{x-1}{1} \Rightarrow y = -x+1}$

$$\mathbf{\therefore \frac{dy}{dx} = -1 + 0 \Rightarrow dy = -dx}$$

Then

$\mathbf{I_2 = \oint(x^2dx - 2xydy)}$ for this part becomes

$$\mathbf{I_2 = \int_1^0 \{x^2dx - 2x(-x+1)(-dx)\} = \int_1^0 (x^2 - 2x^2 + 2x)(dx) = \int_1^0 (2x - x^2)dx = \left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_1^0 = -\frac{2}{3}}$$

c) $\mathbf{BO : c_3}$ is the line $\mathbf{x = 0 \therefore dx = 0}$, Then

$I_3 = \oint(x^2dx - 2xydy)$ for this part becomes

$$I_3 = \oint(0^2.0 - 2.0.ydy) = 0$$

Finally $\mathbf{I = I_1 + I_2 + I_3 = \frac{1}{3} - \frac{2}{3} + 0 = -\frac{1}{3}}$

Q # 95: Evaluate the area of a circle $\mathbf{x^2 + y^2 = 4}$.

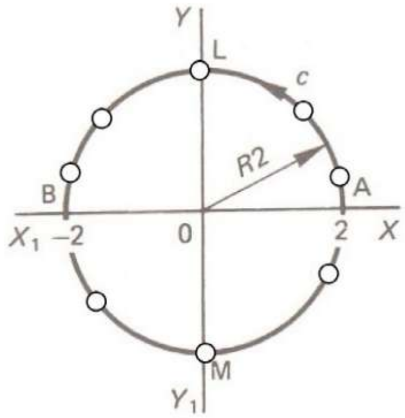

Figure # 127

Answer: Given $\mathbf{x^2 + y^2 = 4}$

$$\mathbf{\Rightarrow y^2 = 4 - x^2}$$

$$\mathbf{\Rightarrow y = \pm\sqrt{4 - x^2}}$$

The Equation of the upper curve ALB: $\mathbf{y = +\sqrt{4 - x^2}}$ between $\mathbf{x = 2}$ and $\mathbf{x = -2}$

And The equation of the lower curve BMA : $\mathbf{y=-\sqrt{4-x^2}}$ between $\mathbf{x=-2}$ and $\mathbf{x=2}$

We have, Area: $A=-\oint y\,dx=-\{\int_{x_1}^{x_2}F(x)dx+\int_{x_2}^{x_1}f(x)dx\}$

$$\therefore A=-[\int_{-2}^{2}\{-(\sqrt{4-x^2})dx+\int_{2}^{-2}\sqrt{4-x^2}dx]$$

$$\therefore A=-[\int_{2}^{-2}\sqrt{4-x^2}dx+\int_{-2}^{2}\{-(\sqrt{4-x^2})dx]$$

$$\therefore A=-[\int_{2}^{-2}\sqrt{4-x^2}dx-\int_{-2}^{2}\sqrt{4-x^2}dx]$$

$$\therefore A=-[\int_{2}^{-2}\sqrt{4-x^2}dx+\int_{2}^{-2}\sqrt{4-x^2}dx]$$

$$\therefore A=-[2\int_{2}^{-2}\sqrt{4-x^2}dx]$$

$$\therefore A=(-2)(-1)\int_{-2}^{2}\sqrt{4-x^2}dx$$

$$\therefore A=2\int_{-2}^{2}\sqrt{4-x^2}dx \text{ ------------------------------(i)}$$

$$\therefore A=2\times2\int_{0}^{2}\sqrt{4-x^2}dx \qquad [\because \int_{-a}^{+a}f(x)dx=2\int_{0}^{a}f(x)dx]$$

$$\therefore A=4\int_{0}^{2}\sqrt{4-x^2}dx \text{ ------------------------------(ii)}$$

Let, $\mathbf{x=2\sin\theta}$

$$\Rightarrow \frac{dx}{d\theta}=\frac{d}{d\theta}(2\sin\theta)$$

$$\Rightarrow \frac{dx}{d\theta}=2\cos\theta$$

$$\Rightarrow dx=2\cos\theta d\theta$$

Now, $\sqrt{4-x^2}=\sqrt{4-(2\sin\theta)^2}=\sqrt{4-4\sin^2\theta}=\sqrt{4(1-\sin^2\theta)}=\sqrt{4\cos^2\theta}=2\cos\theta$

Given, $\mathbf{x=2\sin\theta}$

$$\Rightarrow \theta=\sin^{-1}(x/2)$$

x	0	2
$\theta=\sin^{-1}(x/2)$	$\theta=\sin^{-1}(x/2)$ $\Rightarrow\theta=\sin^{-1}(0/2)$ $\Rightarrow\theta=\sin^{-1}(0)$	$\theta=\sin^{-1}(x/2)$ $\Rightarrow\theta=\sin^{-1}(2/2)$ $\Rightarrow\theta=\sin^{-1}(1)$

	$\Rightarrow \theta = \sin^{-1}\sin 0$ $\Rightarrow \theta = 0$	$\Rightarrow \theta = \sin^{-1}(\sin\frac{\pi}{2})$ $\Rightarrow \theta = \frac{\pi}{2}$

From (ii),

$$\therefore A = 4\int_0^2 \sqrt{4-x^2}\,dx$$

$$\therefore A = 4\int_0^{\pi/2} \sqrt{4-(2\sin\theta)^2}\;2\cos\theta\,d\theta$$

$$\therefore A = 4\int_0^{\pi/2} \sqrt{4-4\sin^2\theta}\;2\cos\theta\,d\theta$$

$$\therefore A = 4\int_0^{\pi/2} \sqrt{4(1-\sin^2\theta)}\;2\cos\theta\,d\theta$$

$$\therefore A = 4\int_0^{\pi/2} \sqrt{4\cos^2\theta}\;2\cos\theta\,d\theta$$

$$\therefore A = 4\int_0^{\pi/2} 2\cos\theta\times 2\cos\theta d\theta$$

$$\therefore A = 4\int_0^{\pi/2} 4\cos^2\theta d\theta$$

$$\therefore A = 4\times 2\int_0^{\pi/2} 2\cos^2\theta d\theta$$

$$\therefore A = 8\int_0^{\pi/2} 2\cos^2\theta d\theta$$

$$\therefore A = 8\int_0^{\pi/2} (1+\cos 2\theta)d\theta \qquad [[2\cos^2\theta = 1+\cos 2\theta]]$$

$$\therefore A = 8\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\pi/2}$$

$$\therefore A = 8\left[(\frac{\pi}{2}-0)+(\frac{\sin 2\times\frac{\pi}{2}}{2}-0)\right]$$

$\therefore A = 8\left[\frac{\pi}{2} + \frac{\sin \pi}{2}\right]$

$\therefore A = 8\left[\frac{\pi}{2} + \frac{0}{2}\right]$

$\therefore A = 8\left[\frac{\pi}{2}\right]$

$\therefore A = 4\pi$ Answer

Or from (i),

$\therefore A = 2\int_{-2}^{2} \sqrt{4 - x^2}\, dx$ ------------------------------(i)

Let, $x = 2\sin\theta$

$\Rightarrow \frac{dx}{d\theta} = \frac{d}{d\theta}(2\sin\theta)$

$\Rightarrow \frac{dx}{d\theta} = 2\cos\theta$

$\Rightarrow dx = 2\cos\theta d\theta$

Now, $\sqrt{4 - x^2} = \sqrt{4 - (2\sin\theta)^2} = \sqrt{4 - 4\sin^2\theta} = \sqrt{4(1 - \sin^2\theta)} = \sqrt{4\cos^2\theta} = 2\cos\theta$

Given, $x = 2\sin\theta$

$\frac{x}{2} = \sin\theta$

$\sin\theta = \frac{x}{2}$

$\Rightarrow \theta = \sin^{-1}(x/2)$

x	-2	2
$\theta = \sin^{-1}(x/2)$	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(-2/2)$ $\Rightarrow \theta = \sin^{-1}(-1)$ $\Rightarrow \theta = -\sin^{-1}(1)$ $\Rightarrow \theta = -\sin^{-1}(\sin\frac{\pi}{2})$ $\Rightarrow \theta = -\frac{\pi}{2}$	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(2/2)$ $\Rightarrow \theta = \sin^{-1}(1)$ $\Rightarrow \theta = \sin^{-1}(\sin\frac{\pi}{2})$ $\Rightarrow \theta = \frac{\pi}{2}$

From (i),

$$\therefore A = 2\int_{-2}^{2} \sqrt{4 - x^2}\, dx$$

$$\therefore A = 2\int_{-\pi/2}^{\pi/2} \sqrt{4 - (2\sin\theta)^2}\ \ 2\cos\theta\, d\theta$$

$$\therefore A = 2\int_{-\pi/2}^{\pi/2} \sqrt{4 - 4\sin^2\theta}\ \ 2\cos\theta\, d\theta$$

$$\therefore A = 2\int_{-\pi/2}^{\pi/2} \sqrt{4(1 - \sin^2\theta)}\ \ 2\cos\theta\, d\theta$$

$$\therefore A = 2\int_{-\pi/2}^{\pi/2} \sqrt{4\cos^2\theta}\ \ 2\cos\theta\, d\theta$$

$$\therefore A = 2\int_{-\pi/2}^{\pi/2} 2\cos\theta \times 2\cos\theta d\theta$$

$$\therefore A = 2\int_{-\pi/2}^{\pi/2} 4\cos^2\theta d\theta$$

$$\therefore A = 2\times 2\int_{-\pi/2}^{\pi/2} 2\cos^2\theta d\theta$$

$$\therefore A = 4\int_{-\pi/2}^{\pi/2} 2\cos^2\theta d\theta$$

$$\therefore A = 4\int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) d\theta \qquad [2\cos^2\theta = 1 + \cos 2\theta]$$

$$\therefore A = 4\int_{-\pi/2}^{\pi/2} 1 d\theta + 4\int_{-\pi/2}^{\pi/2} \cos 2\theta d\theta$$

$$\therefore A = 4[\theta]_{-\pi/2}^{\pi/2} + 4\left[\frac{\sin 2\theta}{2}\right]_{-\pi/2}^{\pi/2}$$

$$\therefore A = 4\left[(\frac{\pi}{2} - (-\pi/2)\right] + 4\left[(\frac{\sin 2\times\frac{\pi}{2}}{2} - \frac{\sin 2\times(-\frac{\pi}{2})}{2})\right]$$

$$\therefore A = 4\left[(\frac{\pi}{2}+\frac{\pi}{2})\right]+4\left[(\frac{\sin\pi}{2}-\frac{\sin(-\pi)}{2})\right]$$

$$\therefore A = 4\left[(\frac{2\pi}{2})\right]+4\left[(\frac{\sin\pi}{2}+\frac{\sin\pi}{2})\right] \qquad [\because \sin(-\theta) = -\sin\theta]$$

$$\therefore A = 4[\pi]+4\left[(\frac{0}{2}+\frac{0}{2})\right] \qquad [\because \sin\pi = 0]$$

$$\therefore A = 4[\pi]+4.0$$

$\therefore A = 4\pi$ Answer

Justification

We know Area of a circle is πr^2

Here radius = r =2

$\therefore$ Area of a circle is $\pi r^2 = \pi\times 2^2 = \pi \times 4 = 4\pi$

Q # 96: If $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$. Evaluate $\iint_s \vec{F}.\hat{\eta}\,ds$

Answer: Given, $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and the scalar function $y^2 = 8x$ .i.e.
$y^2 - 8x = 0$ implies $8x - y^2 = 0$

Let the scalar function $\phi(x, y) = 8x - y^2$ of the given parabolic surface.

We have, $\vec{\nabla}\phi$ is normal (perpendicular) vector to the surface.

Given, $\phi(x, y) = 8x - y^2$

$$\therefore \vec{\nabla}\phi = (\frac{\delta}{\delta x}\hat{i} + \frac{\delta}{\delta y}\hat{j} + \frac{\delta}{\delta z}\hat{k})\phi = (\frac{\delta\phi}{\delta x}\hat{i} + \frac{\delta\phi}{\delta y}\hat{j} + \frac{\delta\phi}{\delta z}\hat{k}) = \hat{i}\frac{\delta}{\delta x}\phi + \hat{j}\frac{\delta}{\delta y}\phi + \hat{k}\frac{\delta}{\delta z}\phi$$

$$= \hat{i}\frac{\delta}{\delta x}(8x - y^2) + \hat{j}\frac{\delta}{\delta y}(8x - y^2) + \hat{k}\frac{\delta}{\delta z}(8x - y^2)$$

$$= 8\hat{i} - 2y\hat{j}$$

Let, $\hat{\eta}$ is the unit vector of $\vec{\nabla}\phi$

We can write,

$$\hat{\eta} = \frac{\vec{\nabla}\phi}{\left|\vec{\nabla}\phi\right|}$$

$$\iint dx\,dy = \int_{\theta=0}^{2\pi}\left[\frac{r^2}{2}-\frac{0^2}{2}\right]d\theta$$

$$\iint dx\,dy = \int_{\theta=0}^{2\pi}\left[\frac{r^2}{2}-0\right]d\theta$$

$$\iint dx\,dy = \frac{r^2}{2}\int_{\theta=0}^{2\pi} d\theta$$

$$\iint dx\,dy = \frac{r^2}{2}\left[\theta\right]_0^{2\pi}$$

$$\iint dx\,dy = \frac{r^2}{2}\left[2\pi-0\right]$$

$$\iint dx\,dy = \frac{r^2}{2}\left[2\pi\right]$$

$$\iint dx\,dy = \pi r^2$$

Q # 109: In three dimensions, $\iiint\limits_V f(x,y,z)\,dx\,dydz$ ----------------------------(i)

Now if we want to switch to another coordinate system, we define $\mathbf{x = x(u,v,w)}$, $\mathbf{y = y(u,v,w)}$ and $\mathbf{z = z(u,v,w)}$

That is: $\iiint\limits_V f(x,y,z)\,dx\,dydz = \iiint\limits_{V'} f(u,v,w)\left|J(u,v,w)\right|du\,dvdw$ ---------------(ii)

Where $\left|J(u,v,w)\right| = \dfrac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} & \dfrac{\partial x}{\partial w} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} & \dfrac{\partial y}{\partial w} \\ \dfrac{\partial z}{\partial u} & \dfrac{\partial z}{\partial v} & \dfrac{\partial z}{\partial w} \end{vmatrix}$

Q# 110: Verify Green's theorem for the integral $\oint\limits_C \{(x^2+y^2)dx+(x+2y)dy\}$ taken round the boundary curve c defined by

$\mathbf{y = 0; 0 \le x \le 2}$

$\mathbf{x^2+y^2 = 4; 0 \le x \le 2}$

$\mathbf{x = 0; 0 \le y \le 2}$

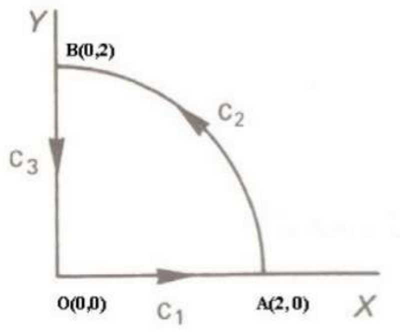

Figure # 161

Answer:

We have Green's theorem $\iint_S \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y}\right) dxdy = \oint_c (Pdx + Qdy)$

LH.S.

First we have to evaluate $\iint_S \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y}\right) dxdy$ ---------------------(i)

Given, $\oint_C \{(x^2 + y^2)dx + (x + 2y)dy\}$ --------------------------------(ii)

Since, R.H.S $\oint_c (Pdx + Qdy)$ ---------------------------------(iii)

Comparing (ii) & (iii),

Here, $P = x^2 + y^2$

$\therefore \frac{\delta P}{\delta y} = 0 + 2y = 2y$

And $Q = x + 2y$

$\therefore \frac{\delta Q}{\delta x} = 1 + 0 = 1$

So, from (i), we get, *[Putting the values of* $\frac{\delta P}{\delta y}$ & $\frac{\delta Q}{\delta x}$ *in (i)]*

$\iint_S \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y}\right) dxdy = \iint_S (1 - 2y)dxdy$ --------------------------(iv)

It will be more convenient to work in polar coordinates, so we make the substitutions

Let $x = r\cos\theta \quad y = r\sin\theta$

Then $\iint_s f(x,y)dxdy = \iint_s f(r,\theta)|J(r,\theta)|\,dr\,d\theta$ --------------------(v)

Here $f(x,y) = 1 - 2y$

From **Jacobian Determinant,**

$\therefore |J(r,\theta)| = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$ ---------------------------------(vi)

We have, $x = r\cos\theta$

$\frac{\partial x}{\partial r} = \frac{\partial}{\partial r}(r\cos\theta)$

$\frac{\partial x}{\partial r} = \cos\theta \frac{\partial}{\partial r}(r)$

$\frac{\partial x}{\partial r} = \cos\theta.1$

$\frac{\partial x}{\partial r} = \cos\theta$ ------------------------------------(vii)

Again

$$\frac{\partial x}{\partial \theta} = \frac{\partial}{\partial \theta}(r\cos\theta)$$

$$\frac{\partial x}{\partial \theta} = r\frac{\partial}{\partial \theta}(\cos\theta)$$

$$\frac{\partial x}{\partial \theta} = -r\sin\theta \qquad \text{--(viii)}$$

Again, $\mathbf{y = r\sin\theta}$

$$\frac{\partial y}{\partial r} = \frac{\partial}{\partial r}(r\sin\theta)$$

$$\frac{\partial y}{\partial r} = \sin\theta\frac{\partial}{\partial r}(r)$$

$$\frac{\partial y}{\partial r} = \sin\theta.1$$

$$\frac{\partial y}{\partial r} = \sin\theta \qquad \text{-------------------------------------(ix)}$$

Again

$$\frac{\partial y}{\partial \theta} = \frac{\partial}{\partial \theta}(r\sin\theta)$$

$$\frac{\partial y}{\partial \theta} = r\frac{\partial}{\partial \theta}(\sin\theta)$$

$$\frac{\partial y}{\partial \theta} = r\cos\theta \qquad \text{--(x)}$$

From (vi),

$$\therefore |\mathbf{J(r,\theta)}| = \frac{\partial(\mathbf{x,y})}{\partial(\mathbf{r,\theta})} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$|\mathbf{J(r,\theta)}| = \frac{\partial(\mathbf{x,y})}{\partial(\mathbf{r,\theta})} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$|\mathbf{J(r,\theta)}| = \frac{\partial(\mathbf{x,y})}{\partial(\mathbf{r,\theta})} = r\cos^2\theta - (-r\sin^2\theta)$$

$$|\mathbf{J(r,\theta)}| = \frac{\partial(\mathbf{x,y})}{\partial(\mathbf{r,\theta})} = r\cos^2\theta + r\sin^2\theta$$

$$|\mathbf{J(r,\theta)}| = \frac{\partial(\mathbf{x,y})}{\partial(\mathbf{r,\theta})} = r(\cos^2\theta + \sin^2\theta)$$

$$|\mathbf{J(r,\theta)}| = \frac{\partial(\mathbf{x,y})}{\partial(\mathbf{r,\theta})} = r.1$$

$$|J(r,\theta)| = \frac{\partial(x,y)}{\partial(r,\theta)} = r \quad \text{---(xi)}$$

From (v),

$$\iint_s f(x,y)dxdy = \iint_s f(r,\theta)|J(r,\theta)|\,dr\,d\theta \quad \text{---------------------------(*)}$$

Hence, From (iv),

$$\iint_S (1-2y)dxdy = \iint_S (1-2r\sin\theta)r\,dr \times d\theta \text{---------------------------(xii)}$$

*[Comparing * & xii,*

[Here, $f(x,y) = 1-2y,\ y = r\sin\theta,\ f(r,\theta) = 1-2r\sin\theta,\ |J(r,\theta)| = r$ *]*

Now,

When $x = 2,\ y = 0$ ***and*** $r = 2$ ***then***

$x = r\cos\theta$

$\Rightarrow 2 = 2\cos\theta$

$\Rightarrow 1 = \cos\theta$

$\Rightarrow \cos 0 = \cos\theta$

$\Rightarrow \theta = 0$

Again

When $x = 0,\ y = 2$ ***and*** $r = 2$ ***then***

$y = r\sin\theta$

$\Rightarrow 2 = 2\sin\theta$

$\Rightarrow 1 = \sin\theta$

$\Rightarrow \sin\frac{\pi}{2} = \sin\theta$

$\Rightarrow \theta = \frac{\pi}{2}$

Hence the limit of θ is $0 \le \theta \le \pi/2$

Again

When $x = 2$ ***and*** $\theta = 0$ ***then***

$x = r\cos\theta$

$2 = r\cos 0$

$\Rightarrow 2 = r \times 1$

$\Rightarrow 2 = r$

$\Rightarrow r = 2$

When $y = 2$ ***and*** $\theta = \pi/2$ ***then***

$y = r\sin\theta$

$2 = r\sin\frac{\pi}{2}$

$\Rightarrow 2 = r \times 1$

$\Rightarrow 2 = r$

$\Rightarrow r = 2$

Hence the limit of $\mathbf{r}$ ***is*** $\mathbf{0 \le r \le 2}$

From (xii),

$$\iint_S (1-2y)\,dxdy = \iint_S (1-2r\sin\theta)\,r\,dr \times d\theta \text{ -}$$

$$= \int_0^{\pi/2}\int_0^2 (1-2r\sin\theta)\,r\,dr\,d\theta$$

$$= \int_0^{\pi/2}\int_0^2 (1-2r\sin\theta)\,r\,dr\,d\theta$$

$$= \int_0^{\pi/2}\int_0^2 (\mathrm{r}-2\mathrm{r}^2\sin\theta)\,\mathrm{d}\mathrm{r}\,\mathrm{d}\theta$$

$$= \int_0^{\pi/2}\left[\frac{r^2}{2} - 2\frac{r^3}{3}\sin\theta\right]_0^2 d\theta$$

$$= \int_0^{\pi/2}\left[\frac{2^2}{2} - 2\frac{2^3}{3}\sin\theta - 0\right] d\theta$$

$$= \int_0^{\pi/2}\left[2 - \frac{16}{3}\sin\theta\right] d\theta$$

$$= \left[2\theta + \frac{16}{3}\cos\theta\right]_0^{\pi/2}$$

$$= \left[2\times\frac{\pi}{2} + \frac{16}{3}\cos\frac{\pi}{2} - 0 - \frac{16}{3}\cos 0\right]$$

$$= \left[\pi + \frac{16}{3}\times 0 - 0 - \frac{16}{3}\times 1\right]$$

$$= \left[\pi - \frac{16}{3}\right]$$

$$\therefore \iint_S \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y}\right) dxdy = \iint_S (1-2y)\,dxdy = \left[\pi - \frac{16}{3}\right] \text{------------------------------(xiii)}$$

Now we have to evaluate R.H.S of Green's Theorem $\oint_c (Pdx + Qdy)$

Given $\oint_C \{(\mathbf{x}^2 + \mathbf{y}^2)\mathbf{dx} + (\mathbf{x} + 2\mathbf{y})\mathbf{dy}\}$

$\therefore \mathbf{P} = \mathbf{x}^2 + \mathbf{y}^2$ and $\mathbf{Q} = \mathbf{x} + 2\mathbf{y}$

We now take $\mathbf{c}_1$, $\mathbf{c}_2$, $\mathbf{c}_3$ in turn

i) $\mathbf{c}_1 : \mathbf{y} = \mathbf{0};\ \therefore \mathbf{dy} = \mathbf{0}$

$$\therefore \oint_{c_1}(Pdx + Qdy) = \int_0^2 (x^2 + y^2)dx + (x + 2y)dy = \int_0^2 (x^2 + 0^2)dx + (x + 2.0).0$$

$$= \int_0^2 x^2 dx = \frac{8}{3}$$

ii) $c_2 : x^2 + y^2 = 4 \therefore y^2 = 4 - x^2 \therefore y = \sqrt{4 - x^2} = (4 - x^2)^{1/2}$

$$\therefore \frac{dy}{dx} = \frac{1}{2}(4 - x^2)^{-1/2}.\frac{d}{dx}(4 - x^2)$$

$$\therefore \frac{dy}{dx} = \frac{1}{2}(4 - x^2)^{-1/2}.(0 - 2x)$$

$$\therefore dy = -(4 - x^2)^{-1/2}.xdx = \frac{-xdx}{\sqrt{4 - x^2}}$$

$$\therefore \oint_{c_2}(Pdx + Qdy) = \oint_{c_2}(x^2 + y^2)dx + (x + 2y)dy$$

$$= \oint_{c_2}(x^2 + 4 - x^2)dx + (x + 2\sqrt{4 - x^2}).\frac{-xdx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2} 4dx - (x + 2\sqrt{4 - x^2})\frac{xdx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2} 4dx - \frac{x^2dx}{\sqrt{4 - x^2}} - 2\sqrt{4 - x^2}\frac{xdx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2} 4dx - \frac{x^2dx}{\sqrt{4 - x^2}} - 2xdx$$

$$= \oint_{c_2}(4 - 2x)dx - \frac{x^2dx}{\sqrt{4 - x^2}}$$

$$= \oint_{c_2}(4 - 2x - \frac{x^2}{\sqrt{4 - x^2}})dx$$

$$= \int_2^0 (4 - 2x - \frac{x^2}{\sqrt{4 - x^2}})dx \text{ -------------------------------------(xiv)}$$

Let $x = 2\sin\theta$

$\Rightarrow dx = 2\cos\theta d\theta$

$$\therefore \sqrt{4 - x^2} = \sqrt{4 - (2\sin\theta)^2} = \sqrt{4 - 4\sin^2\theta} = \sqrt{4(1 - \sin^2\theta)} = \sqrt{4\cos^2\theta} = 2\cos\theta$$

Given, $x = 2\sin\theta$

$\Rightarrow \theta = \sin^{-1}(x/2)$

x	0	2
θ	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(0/2)$ $\Rightarrow \theta = \sin^{-1}(0)$ $\Rightarrow \theta = \sin^{-1}\sin 0$ $\Rightarrow \theta = 0$	$\theta = \sin^{-1}(x/2)$ $\Rightarrow \theta = \sin^{-1}(2/2)$ $\Rightarrow \theta = \sin^{-1}(1)$ $\Rightarrow \theta = \sin^{-1}(\sin\frac{\pi}{2})$ $\Rightarrow \theta = \frac{\pi}{2}$

$$\therefore \oint_{c_2}(Pdx + Qdy) = \int_2^0 (4 - 2x - \frac{x^2}{\sqrt{4-x^2}})dx$$

$$= \int_{\pi/2}^0 \{(4 - 2\times 2\sin\theta - \frac{(2\sin\theta)^2}{2\cos\theta}\}2\cos\theta d\theta$$

$$= \int_{\pi/2}^0 \{(8\cos\theta - 4\times 2\sin\theta\cos\theta - 4\sin^2\theta)d\theta$$

$$\therefore \oint_{c_2}(Pdx + Qdy) = \int_{\pi/2}^0 8\cos\theta d\theta - \int_{\pi/2}^0 4\times 2\sin\theta\cos\theta d\theta - \int_{\pi/2}^0 4\sin^2\theta d\theta \text{ --------------(xv)}$$

Now,

a) $\int_{\pi/2}^0 8\cos\theta d\theta = [8\sin\theta]_{\pi/2}^0 = 8\sin 0 - 8\sin\frac{\pi}{2} = 0 - 8\times 1 = -8$

b) $\int_{\pi/2}^0 4\times 2\sin\theta\cos\theta d\theta$

Let $z = \sin\theta$

$\Rightarrow dz = \cos\theta d\theta$

Since $z = \sin\theta$

$\Rightarrow \theta = \sin^{-1}(z/2)$

θ	0	$\frac{\pi}{2}$
z	$z = \sin\theta$ $z = \sin 0$ $z = 0$	$z = \sin\theta$ $z = \sin \pi/2$ $z = 1$

$$\int_{\pi/2}^0 4\times 2\sin\theta\cos\theta d\theta = \int_1^0 4\times 2z dz = \left[8\times\frac{z^2}{2}\right]_1^0 = 4(0-1) = -4$$

c) $\int_{\pi/2}^{0} 4\sin^2\theta d\theta = 2\int_{\pi/2}^{0} 2\sin^2\theta d\theta = 2\int_{\pi/2}^{0}(1+\cos 2\theta)d\theta$

$$= 2\left[\theta - \frac{\sin 2\theta}{2}\right]_{\pi/2}^{0} = 2(0-\frac{1}{2}\sin 2.0 - \frac{\pi}{2} + \frac{1}{2}\sin 2.\frac{\pi}{2}) = 2(0-0-\frac{\pi}{2}+\frac{1}{2}.\sin\pi) = 2(-\frac{\pi}{2}+\frac{1}{2}.0)$$

$= -\pi$

From (xv)

$$\therefore \oint_{c_2}(Pdx + Qdy) = \int_{\pi/2}^{0} 8\cos\theta d\theta - \int_{\pi/2}^{0} 4\times 2\sin\theta\cos\theta d\theta - \int_{\pi/2}^{0} 4\sin^2\theta d\theta$$

$$= -8 - (-4) - (-\pi)$$

$$= -8 + 4 + \pi$$

$$= \pi - 4$$

iii) $c_3 : x = 0;\ \therefore dx = 0$

$\therefore \oint_{c_3}(Pdx + Qdy)$

$= \oint_{c_2}(x^2 + y^2)dx + (x + 2y)dy$

$= \oint_{c_3}(0^2 + y^2).0 + (0 + 2y)dy = \int_2^0 2ydy = \left[2\frac{y^2}{2}\right]_2^0 = -4$

Collecting our three partial results

$\oint_c(Pdx + Qdy) = \frac{8}{3} + \pi - 4 - 4 = \pi - \frac{16}{3}$ --------------------(xvi)

From (xiii) and (xvi), we can write

L.H.S. = R.H.S.

$\therefore \iint_S\left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y}\right)dxdy = \oint_c(Pdx + Qdy)$ ***(Proved)***

Q# 111: Use Green's theorem to evaluate $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$ where C is the square formed ABCD by the lines $y = \pm 1, x = \pm 1$

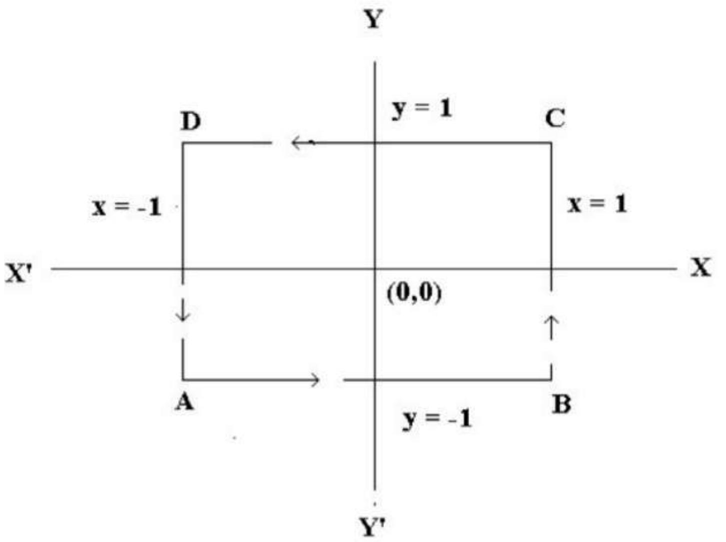

Figure # 162

Answer: From Green's theorem, we have,

$$\int_C Pdx + Qdy = \iint_R \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y}\right)dxdy$$

Here, $P = x^2 + xy$ and $Q = x^2 + y^2$

$$\int_C Pdx + Qdy = \iint_R \left(\frac{\delta Q}{\delta x} - \frac{\delta P}{\delta y}\right)dxdy$$

L.H.S.

$\int_C Pdx + Qdy =$ TRY YOURSELF

Here, R.H.S= $\int_{-1}^{1}\int_{-1}^{1} \{(\frac{\delta}{\delta x}(x^2 + y^2) - \frac{\delta}{\delta y}(x^2 + xy)\}dxdy$

$$= \int_{-1}^{1}\int_{-1}^{1} (2x - x)dxdy$$

$$= \int_{-1}^{1}\int_{-1}^{1} xdxdy$$

$$= \int_{-1}^{1}\left[\frac{x^2}{2}\right]_{-1}^{1} dy$$

$$= \int_{-1}^{1}\left[\frac{1^2}{2} - \frac{(-1)^2}{2}\right]dy$$

$$= \int_{-1}^{1}\left[\frac{1^2}{2} - \frac{1^2}{2}\right]dy$$

$= \mathbf{0}$ Answer

Home Task:

State Green's theorem. Verify Green's theorem in the plane for

$\int_C (2xy - x^2)dx + (x + y^2)dy$ where the close curve of the region bounded by

$y = x, y^2 = x$

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